CITNet: Convolution Interaction Transformer Network for Hyperspectral and LiDAR Image Classification

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***Abstract*— Transformers are increasingly popular in computer vision, which treat an image as a sequence of image patches and learn robust global features from the sequence. However, pure transformers are not entirely suitable for hyperspectral and light detection and ranging (LiDAR) image classification because image classification requires both robust global features and discriminative local features. Therefore, this article introduces a novel convolution interaction transformer network (CITNet) for jointly classifying hyperspectral and LiDAR images. The process begins with a carefully designed multiscale asymmetric depthwise convolution (MADC) module that exploits the local–global correlations of shallow features. On this basis, a novel local–global transformer (LGTM) is equipped with a local–global feed-forward (LGF) network to extract in-depth local–global joint features from the multimodal data. Then, an optimization convolution cross-attention (OCA) module, incorporating a convolutional layer, is developed to simulate the spatial relationships of semantic tokens. Finally, extensive experiments are conducted on the well-known Trento (TR), Augsburg (AU), MUUFL (MU), and Houston2013 (HU) datasets. The overall accuracy (OA) reaches 99.76%, 97.40%, 91.06%, and 99.90%, respectively, which are 0.2%–1.66%, 0.32%–7.37%, 1.52%–12.71%, and 0.14%–93.79% higher than the state-of-the-art (SOTA) methods, demonstrating the effectiveness of CITNet in improving the joint classification accuracy of hyperspectral and LiDAR images.**

***Index Terms*— Convolutional neural network (CNN), cross attention (CA), hyperspectral image (HSI), light detection and ranging data, vision transformer.**

I. Introduction

DATA from various remote sensing imaging devices focused on the same geographic area are accessible, enabling the analysis of land cover material using multimodal data [1], [2]. Different remote sensing imaging sensors excel in capturing distinct attributes of land cover material. Hyperspectral imagers, for instance, gather reflected spectral information alongside spatial details of the terrain [3]. At the same time, light detection and ranging (LiDAR) systems ascertain the elevation details of terrestrial objects [4]. A more detailed and complete feature representation of the land objects can be achieved by amalgamating multimodal data [5].

Hyperspectral imagers have found extensive applications in areas such as land-cover classification [6], [7], ecosystem monitoring [8], anomaly detection [9], [10], [11], [12], and among others [13], [14], [15], [16]. Notably, land cover classification, often referred to as hyperspectral image classification (HSIC), stands out as a critical task. The HSIC task mainly uses single-modality data, which uses the spectral–spatial dimension information of the hyperspectral image (HSI) to assign a category identifier to each pixel. With the advancement of deep learning [17], [18], [19], [20], novel methods have substantially enhanced classification performance [21], [22]. For example, the HybridSN [23] combines the 3-D convolutional neural network (CNN) with the 2-D CNN to improve spectral–spatial feature representation. The SpectralFormer [24] utilizes the self-attention mechanism (SA) to capture complex dependencies within the features. The spectral–spatial feature tokenization transformer (SSFTT) [25] encapsulates token relationships and extracts high-level semantic features.

While HSI alone has achieved promising results, misclassification is inevitable due to the spectral feature variations between the same ground objects and the appearance of similar spectral features [26] in different ground objects. LiDAR-derived digital surface models (LiDAR-DSMs) data provide insights into terrain variations and the height of surface object features, which are often combined with HSI for joint classification to improve accuracy. Numerous studies have explored the synergistic integration of HSI and LiDAR-DSM data. Early methods primarily centered on joint feature extraction using morphological profiles [27]. However, directly merging high-dimensional features can induce the Hughes phenomenon. To address this issue, techniques such as principal component analysis (PCA) [28] and total variation component analysis (TVCA) [29] were introduced to reduce dimensionality and enhance feature extraction. Traditional

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methods primarily integrate prior knowledge, which lays not only a foundation for classification decisions but also imposes constraints that could hinder advancements in accuracy.

The emergence of deep learning has significantly solved the deficiencies of traditional methods. For instance, the hash-based deep metric learning approach [30] emphasizes sample correlations within both single-source and cross-source data. The straightforward dual-branch coupled CNN [31] independently extracts features from HSI and LiDAR-DSM data. The hierarchical random walk network (HRWN) [32] combines dual-channel feature extraction with the hierarchical random walk technique. Inspired by HSIC, recent joint classification studies of HSI and LiDAR-DSM data have numerous CNN-Transformer methods, showing remarkable efficacy. The global–local transformer network (GLT-Net) [33] captures the global–local features by concatenating features from HSI and LiDAR-DSM. The local information interaction transformer (LIIT) [34] dynamically integrates multimodal features to tackle the challenges associated with redundant or insufficient complementary information. The multibranch interaction transformer [35] simultaneously extracts spectral, spatial, and elevation information. The multimodal fusion transformer (MFT) [36] derives feature representations from multiple data sources, utilizing the class token for final classification. The hierarchical CNN and Transformer (HCT) [37] fuses multisource heterogeneous information to enhance joint classification efficiency. However, existing approaches often neglect local–global dependencies, focus on superposition without effective interaction, or limit interaction to shallow layers. Thus, it is worthwhile to consider designing a novel CNN-Transformer approach to systematically extract joint local–global features from shallow to deep layers while integrating multimodal data.

To comprehensively extract the spectral–spatial features of HSI and the elevation features of LiDAR-DSM, address issues overlooked by existing joint methods, and enhance classification accuracy, a novel convolution interaction transformer network (CITNet) is designed. The model captures long-range dependencies to extract local–global joint features of multimodal data from shallow-to-deep layers. Several crucial innovations contribute to its uniqueness.

1) A multiscale asymmetric depthwise convolutional (MADC) information extraction module is designed to enhance local–global feature interaction at a shallow level by utilizing small and large convolutional kernels, effectively improving classification performance.
2) An innovative local–global transformer (LGTM) with a local–global feed-forward (LGF) network is proposed to fully leverage local and global properties within semantic tokens, enriching feature representation by interacting spatial and channel dimensions.
3) An optimization convolution cross attention (OCA) is designed to achieve enhanced accuracy and robustness in classification. It effectively establishes the spatial relationships among semantic tokens, significantly facilitating the integration of local and global features.

The remainder of this article is structured as follows. Section II covers related works. Section III details the proposed CITNet. Section IV presents the datasets, experimental settings, and results. Section V conducts ablation studies, evaluates the performance of different components, validates the superiority of multimodal image classification over HSIC, and verifies model robustness across different training samples. Section VI concludes this article.

II. Related Works

A. Multimodal Data Fusion

The advancement of sensor technology has led to a growing emphasis on the judicious utilization of multimodal data, making it a focal point of research on remote sensing. The multimodal transfer feature fusion network (MTFFN) [38] uses a dual-branch framework for natural images and HSI, where each branch independently predicts ground object classes. While it balances modality-specific features, it lacks deeper integration. The enhanced U-Net (E-UNet) [39] learns deep shared features from multimodal remote sensing images to improve change detection. Although effective at feature capture, the complexity of the feature fusion process may constrain its performance. Collaborative contrastive learning (CCL) [40] addresses the challenge of limited labeled samples for HSI and LiDAR data classification but may struggle with generalization when the available data are highly imbalanced across modalities. The Gaussian mixture variational dynamic fusion network (GM-VDFN) [41] learns adaptive fusion patterns across different modalities, offering flexibility in handling diverse data sources. However, it has high computational complexity. Extended vision transformer (ExViT) [42] extended with separable convolutional blocks, which can use spatial and modality-specific channel information together. This method enhances the utilization of multimodal features but might be less effective when the modalities have significantly different spatial resolutions or noise levels. The shuffle feature enhancement-based fusion network (SFE-FN) [43] addresses the noise issue in multimodal data fusion but may increase complexity in feature extraction.

These studies highlight the strengths of multimodal data fusion in enhancing feature representation to improve classification accuracy. However, challenges remain, including computational complexity and the need for balanced data across modalities.

B. Multiscale-Based Methods

The incorporation of multiscale features in image analysis enables the extraction of more comprehensive information. Small scales facilitate the acquisition of detailed local information, while larger scales capture the overall global features. Numerous studies have explored remote sensing image processing and analysis by multiscale features. The multiscale deep neural network with a hierarchical residual architecture and self-calibrated convolution (MSNetSC) [44] enhances feature extraction across diverse receptive fields, capturing fine and coarse features. However, its hierarchical structure increases computational costs. The heterogeneous spectral–spatial network (HSSAM) [45] integrates a 3-D multiscale dense residual asymmetric convolution module

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(3D-MRDAC) and a 3-D attention mechanism, effectively enhancing feature extraction and classification performance. However, its superior feature representation comes with high computational complexity. The dynamic scale hierarchical fusion network (DSHFNet) [46] uses various attention modules for hierarchical fusion, effectively integrating multiscale information. While it enhances feature complementarity, the complexity of its attention mechanisms may complicate training and optimization. The multiscale pyramid fusion framework (MS2CANet) [47] uses spatial–spectral cross-modal attention (S2CA) to learn multiscale information in regions with complex changes. However, it may be prone to overfitting on datasets with limited samples. The multiscale head selection transformer network (MHST) [48] learns detailed and nonredundant feature information across spatial and spectral dimensions of HSI and LiDAR-DSM data, effectively balancing detail capture and redundancy reduction. However, its reliance on transformer architecture may result in significant memory demands.

These methods demonstrate the effectiveness of multiscale approaches in capturing both local and global features, improving accuracy and robustness in remote sensing tasks. However, cascading multiscale features can significantly increase dimensionality, requiring careful design to prevent excessive feature dimensions from negatively impacting computational efficiency and model performance.

C. CNN-Transformer-Based Methods

Deep learning methods have significantly advanced remote sensing representation. The fractional Gabor convolutional network (FGCN) [49] uses the octave and fractional Gabor convolutions to cut down redundant features and take multiscale and multidirectional spectral–spatial features. While the SpectralFormer [24] leverages the Vision Transformer to extract discriminative spectral features from HSIs. Despite their strengths, CNNs struggle with long-sequence spectral features and transformers, though effective in modeling global interactions, are often weak in capturing local information.

Recent hybrid models such as the SSFTT [25] and the GLT-Net [33] aim to combine the strengths of CNNs and transformers. SSFTT integrates a CNN with a transformer to enhance sequential and semantic feature extraction, while GLT-Net captures global–local features but lacks effective multimodal interactive learning. The LIIT [34] and the learnable transformer with an adaptive gating mechanism (AGMLT) [50] attempt to integrate multimodal features dynamically. The former faces challenges in capturing the dependencies between local and global features, and the latter has high model complexity. The MFT [36] and HCT [37] operate on semantic tokens. MFT directly utilizes these tokens for classification, whereas HCT reduces model complexity by compressing the tokens. However, both approaches are short in capturing dependencies between local and global features.

Therefore, it is necessary to explore a CNN-transformer approach that effectively extracts and fuses local and global features from shallow-to-deep layers and enables comprehensive multimodal feature integration.

III. Methodology

The proposed CITNet is illustrated in Fig. 1. First, the MADC is designed to enhance the extraction of shallow local–global features. Then, the LGTM with LGF is proposed to obtain local–global dependencies. Whereafter, OCA with convolution is designed to fuse multimodal data. The original input of HSI and LiDAR-DSM could be represented as $\boldsymbol{X}_{\text{ori}}^{\text{H}} \in R_{H\times W\times B}$ and $\boldsymbol{X}_{\text{ori}}^{\text{L}} \in R_{H\times W}$, where H and W denote the spatial size and B is the number of spectra for HSI. HSI encompasses a large number of spectral bands that not only deliver rich information but also substantially elevate computational costs. To decrease the spectral dimension, the PCA is adopted to extract the first C principal components of HSI, and the HSI after PCA is $\boldsymbol{X}_{\text{PCA}}^{\text{H}} \in R_{H\times W\times C}$. On this basis, with each spatial position (i, j) as the center, here $i \in \{1, \ldots, H\}$, and $j \in \{1, \ldots, W\}$. The image patches $\boldsymbol{X}_{\text{in}}^{\text{H}} \in R_{P\times P\times C}$ and $\boldsymbol{X}_{\text{in}}^{\text{L}} \in R_{P\times P}$ with the same spatial size are extracted from $\boldsymbol{X}_{\text{PCA}}^{\text{H}}$ and $\boldsymbol{X}_{\text{ori}}^{\text{L}}$, respectively, where $P \times P$ represents the spatial size of the image patches [51]. Divide the image patches that are not 0 labels into training sets $\boldsymbol{D}_{\text{train}}$ and testing sets $\boldsymbol{D}_{\text{test}}$, where $\boldsymbol{D}_{\text{train}}$ contains the class labels and $\boldsymbol{D}_{\text{test}}$ does not contain the class labels.

A. Feature Extraction via MADC

While transformer networks excel at simulating global interactions between tokens by SA, their ability to extract fine-grained local feature patterns is somewhat limited [52]. In contrast, CNN is particularly adept at modeling spatial context features, which enables them to excel in tasks such as HSIC. Their effectiveness is further demonstrated in applications where CNN extracts deep features from LiDAR-DSM data. Motivated by these capabilities, we incorporate CNN to enhance shallow feature extraction from our input data. As shown in Fig. 1, a dual-branch MADC module is designed to extract shallow local–global features from HSI and LiDAR-DSM. MADC has sub-branches of different convolutional kernels (i.e., 1, 5, and 11). The sub-branches with a convolutional kernel size of 1 serve as residual connections. These 1×1 convolutions align the feature dimensions, as the output dimensions of other branches differ from the original input. This alignment ensures that features from different branches can be superimposed while preserving the original information. The sub-branches with the small convolutional kernel size 5 focus on local feature extraction. The sub-branches with the large convolutional kernel size 11 focus on global feature extraction.

Since HSI is 3-D data, according to the HybridSN [23], the 3DCNN effectively captures spectral–spatial joint information but is often computationally expensive, and the 2DCNN is more computationally efficient but is more suitable for extracting spatial features. Combining the two can achieve a more efficient and comprehensive feature extraction process. In addition, since the HSI shallow feature extraction method of the baseline method HCT [37] is 3-D–2-D CNN, to ensure the fairness of the experiment, this article selects 3-D–2-D MADC for HSI feature extraction. Due to LiDAR-DSM being 2-D data, it is processed by the 2-D MADC. The local

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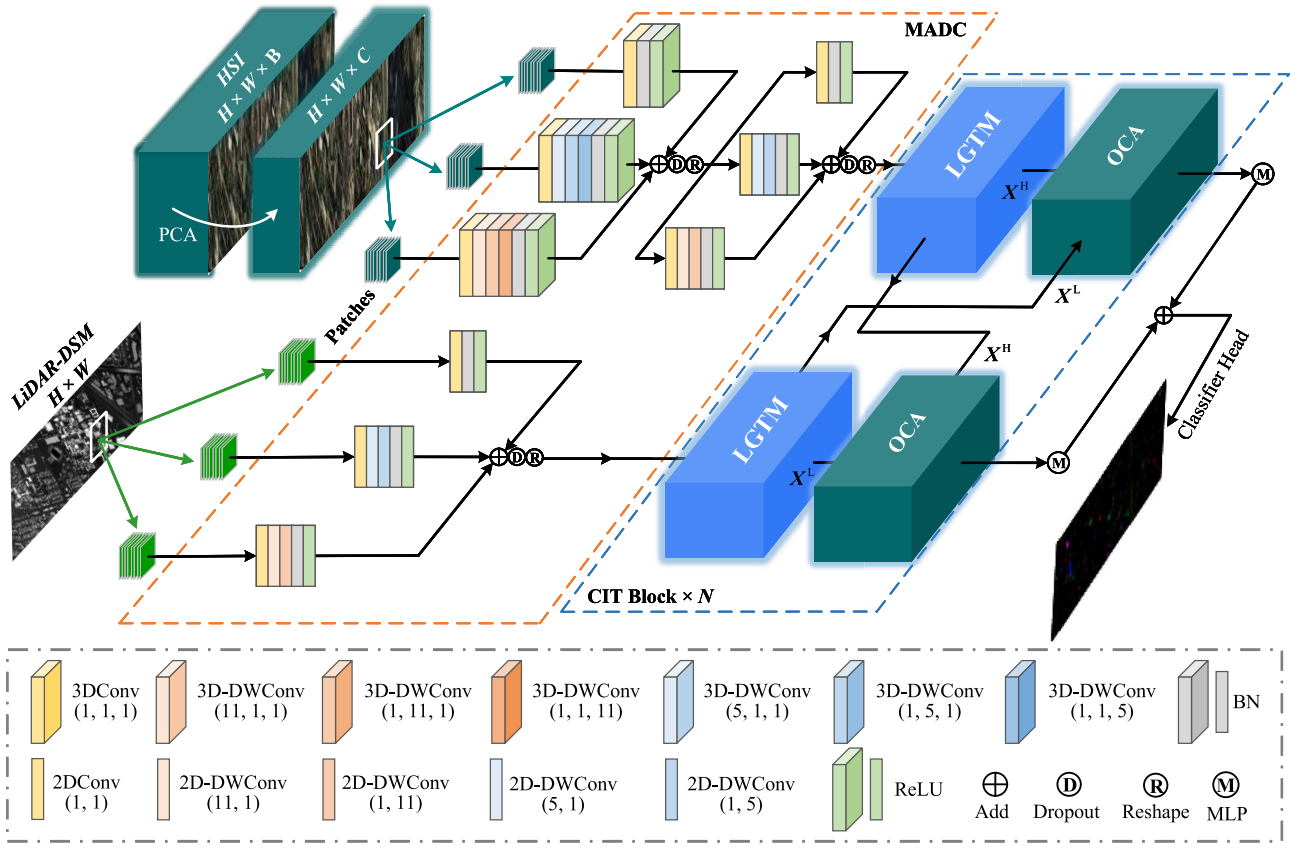


Fig. 1. Illustration of the proposed CITNet. (The CITNet can be divided into two parts. The one part is the MADC module, which can extract the shallow local–global features. The other part is the loopable CIT block, which contains the LGTMs and the OCAs. LGTM is designed to extract local–global joint features of semantic tokens. OCA is proposed for modeling spatial information of semantic tokens and fusing multimodal features. In this figure, $H \times W \times B$ represents the size of HSI raw data. $H \times W$ represents the size of LiDAR-DSM raw data. $H \times W \times C$ represents the size of the HSI processed by PCA. $\boldsymbol{X}^{\mathrm{H}}$ is the semantic tokens from LGTM in the HSI branch, and $\boldsymbol{X}^{\mathrm{L}}$ is the semantic tokens from LGTM in the LiDAR-DSM branch.)

feature extraction sub-branches and the global feature extraction sub-branches all adopt asymmetric convolution (AC) [53] and depthwise convolution (DWC) [54] to form the asymmetric depthwise convolution (ADC). The principle is that when a 2-D kernel is of rank 1, it can effectively be decomposed into a sequence of 1-D convolutions, which could enhance the core structure of the CNN and reduce the number of parameters [53]. When each dimension of a 3DCNN kernel is 1, respectively, it essentially functions as a 2DCNN to extract spatial relationships between different dimensions of HSI. Therefore, the 3DCNN can employ AC kernels. Moreover, each kernel in DWC is allocated to different feature channels, which can significantly reduce the number of parameters and computational costs compared with standard CNN [54].

The 3-D–2-D MADC for HSI is in the top left corner of Fig. 1. The standard convolutional kernels in the 3-D MADC are $1 \times 1 \times 1$, the ADC kernels of the local feature extraction sub-branch in 3-D MADC are $5 \times 1 \times 1$, $1 \times 5 \times 1$, and $1 \times 1 \times 5$, and the ADC kernels of the global feature extraction sub-branch in 3-D MADC are $11 \times 1 \times 1$, $1 \times 11 \times 1$, and $1 \times 1 \times 11$. The outputs of the 3-D residual sub-branch, 3-D local feature extraction sub-branch, and 3-D global feature extraction sub-branch are superimposed. The 3-D overlay outputs are reshaped into 2-D into 2-D features after the dropout. Standard convolutional kernels in the 2-D MADC are 1×1, the ADC kernels of the local feature extraction sub-branch in 2-D MADC are 5×1 and 1×5, and the ADC kernels of the global feature extraction sub-branch in 2-D MADC are 11×1 and 1×11. The outputs of the 2-D residual sub-branch, the 2-D local feature extraction sub-branch, and the 2-D global feature extraction sub-branch are also superimposed. After the dropout, 2-D overlay outputs are reshaped to the 1-D vector to match the subsequent LGTM. The 2-D MADC for LiDAR-DSM is in the left bottom of Fig. 1, whose operations are consistent with the 2-D parts in the 3-D–2-D MADC for HSI.

B. LGTM Encoder

Fig. 2 shows the detailed structure of the proposed LGTM encoder. We make improvements based on the baseline HCT. Fig. 2(a) shows the transformer encoder in HCT, and Fig. 2(b) shows the proposed encoder LGTM. The multilayer perceptron (MLP) feed-forward network (FFN) is not skilled in establishing the relationship between channel and spatial features. Directly replacing MLP with regular convolutional operations may also lead to increased computational cost. Therefore, inspired by LightViT [55], a new local–global FFN named LGF shown in Fig. 3 is designed.

First, through the tokenization [37] operation, the deep-level semantic information of shallow local–global features is mined. The convolutional feature maps are flattened as the

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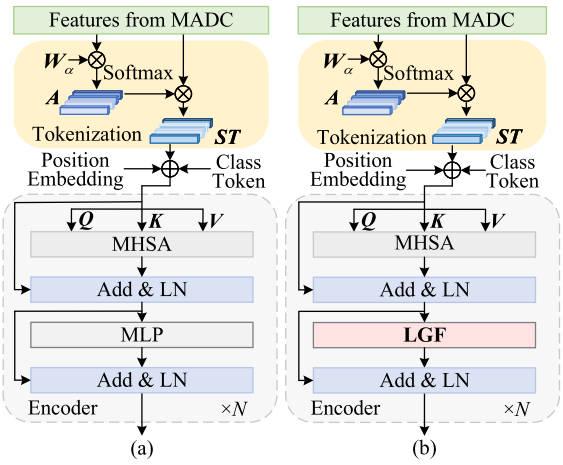


Fig. 2. Architecture of transformer encoder. (a) Transformer in HCT [37]. (b) Our LGTM. (Tokenization [37] redefines the obtained shallow local–global features as semantic tokens. LGF is a novel LGF network. $\boldsymbol{W}_\alpha$ is the learnable weight matrices persuaded by Gaussian distribution. $\boldsymbol{A}$ is the generated semantic group. $\boldsymbol{ST}$ is the semantic tokens after Tokenization. $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are the query, key, and value matrices, respectively. Add is the residual connection and LN is the layer normalization. $\otimes$ represents the multiplication, and $\oplus$ stands for adding the class token and position tokens to semantic tokens. The encoder can cycle N times.)

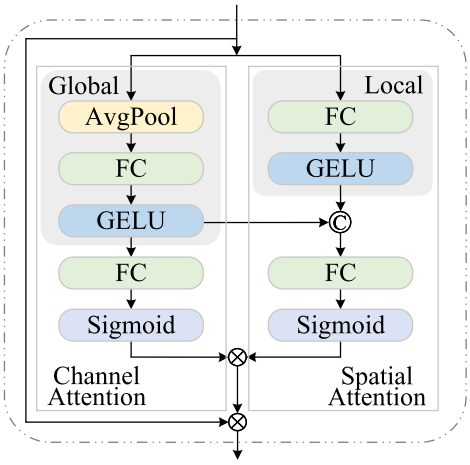


Fig. 3. Architecture of LGF. (FC is the Linear(·), GELU is the Gaussian error linear unit, $\otimes$ is the multiplication, and © is the concat.)

input and expressed as $\boldsymbol{X}^{\mathrm{F}} \in R^{hw\times c}$, where h is the height, w is the width, and c is the number of channels. The semantic tokens are defined as $\boldsymbol{ST} \in R^{t\times c}$ where t is the number of semantic tokens. For $\boldsymbol{X}^{\mathrm{F}}$, $\boldsymbol{ST}$ can be expressed as follows:

$$\boldsymbol{ST} = \underbrace{\mathrm{Softmax}\left(\boldsymbol{X}^{\mathrm{F}}\boldsymbol{W}_\alpha\right)}_{\boldsymbol{A}}{}^{\mathrm{T}}\boldsymbol{X}^{\mathrm{F}} \tag{1}$$

where $\boldsymbol{W}_\alpha \in R^{c\times t}$ is the learnable weight matrices persuaded by Gaussian distribution, and $\boldsymbol{X}^{\mathrm{F}}\boldsymbol{W}_\alpha$ represents the point-by-point multiplication of them. Then, $\boldsymbol{X}^{\mathrm{F}}$ is generated to semantic group $\boldsymbol{A} \in R^{hw\times t}$ by the Softmax(·). Transposed $\boldsymbol{A}$ is multiplied by $\boldsymbol{X}^{\mathrm{F}}$, and the weighted average of the feature pixels in the $\boldsymbol{X}^{\mathrm{F}}$ is calculated to obtain t semantic tokens. The operations of the HSI branch and the LiDAR-DSM branch are the same.

The transformer encoder can capture relevant information between semantic tokens, which reduces dependence on external information. A learnable class token T_0^{cls} is applied to the first position of the semantic tokens, while the positional embedding is carried out to the semantic tokens. The new holistic semantic token is $\boldsymbol{ST}' \in R^{(t+1)\times c}$. To learn the information within the holistic semantic token, learnable weights $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$ are predefined for SA. Multiply the holistic semantic token with $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$, and linearly package them to query, key, and value ($\boldsymbol{Q}$, $\boldsymbol{K}$, $\boldsymbol{V}$) matrices. The Softmax(·) shifts the scores into weight probabilities. SA is written as follows:

$$\mathrm{SA}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\mathrm{T}}}{\sqrt{d_K}}\right)\boldsymbol{V} \tag{2}$$

where d_K is the dimension of $\boldsymbol{K}$ matrix. The multihead SA (MHSA) has multiple SA, and all scores of SA are tied together, whose expression is

$$\mathrm{MHSA}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \mathrm{Concat}(\mathrm{SA}_{h1},\mathrm{SA}_{h2},\ldots,\mathrm{SA}_{hn})\boldsymbol{W} \tag{3}$$

where h_n is the number of attention heads and $\boldsymbol{W}$ is the parameter matrix.

Put the learned holistic semantic token $\boldsymbol{ST}'' \in R^{(t+1)\times c}$ after layer normalization (LN) and residual connection into LGF to enhance the feature representation ability of CITNet. The channel attention (CHA) branch utilizes average pooling on the spatial dimensions of features to encapsulate global attributes and leverages linear transformations for CHA computation. Concurrently, the spatial attention (SPA) branch uses linear transformations for SPA assessment and merges this with global representations, forming pixel-specific relationships that illustrate local features. Therefore, LGTM can extract in-depth local–global semantic features, as shown in the below formulas:

$$\boldsymbol{ST}_{\mathrm{lo}} = \mathrm{GELU}\left(\mathrm{Linear}\left(\boldsymbol{ST}''\right)\right) \tag{4}$$

$$\boldsymbol{ST}'_{\mathrm{gl}} = \mathrm{Avgpool}\left(\boldsymbol{ST}''\right) \tag{5}$$

$$\boldsymbol{ST}_{\mathrm{gl}} = \mathrm{GELU}\left(\mathrm{Linear}\left(\boldsymbol{ST}'_{\mathrm{gl}}\right)\right) \tag{6}$$

where $\boldsymbol{ST}_{\mathrm{lo}} \in R^{(t+1)\times c_{\mathrm{lo}}}$ is the local semantic token. $t+1$ represents the number of semantic tokens equipped with the class token. c_{lo} is the channel dimension of local semantic tokens. $\boldsymbol{ST}'_{\mathrm{gl}} \in R^{1\times c}$ is a global semantic token, whose channel dimension is the same as $\boldsymbol{ST}''$, and information of all semantic tokens are pooled into one semantic token. $\boldsymbol{ST}_{\mathrm{gl}} \in R^{1\times c_{\mathrm{lo}}}$ is a final global semantic token with the same channel dimension with local semantic tokens.

The fusion of spatial dimension and channel dimension is shown in the following formulas:

$$\boldsymbol{ST}_{\mathrm{SPA}} = \mathrm{Sigmoid}\left(\mathrm{Linear}\left(\mathrm{Concat}\left(\boldsymbol{ST}_{\mathrm{lo}},\boldsymbol{ST}_{\mathrm{gl}}\right)\right)\right) \tag{7}$$

$$\boldsymbol{ST}_{\mathrm{CHA}} = \mathrm{Sigmoid}\left(\mathrm{Linear}\left(\boldsymbol{ST}_{\mathrm{gl}}\right)\right) \tag{8}$$

$$\boldsymbol{ST}^{\mathrm{out}}_{\mathrm{LGF}} = \left(\boldsymbol{ST}_{\mathrm{SPA}}\boldsymbol{ST}_{\mathrm{CHA}}\right)\boldsymbol{ST}'' \tag{9}$$

where $\boldsymbol{ST}_{\mathrm{SPA}} \in R^{(t+1)\times 1}$ is the channel semantic token with the one channel dimension, which focuses on the relationship between semantic tokens. $\boldsymbol{ST}_{\mathrm{CHA}} \in R^{1\times c}$ focus on the channel information. $\boldsymbol{ST}_{\mathrm{SPA}}\boldsymbol{ST}_{\mathrm{CHA}}$ fuses channel and spatial semantic information by multiplying point by point and then merges the original semantic information by multiplying with $\boldsymbol{ST}''$. $\boldsymbol{ST}^{\mathrm{out}}_{\mathrm{LGF}} \in R^{(t+1)\times c}$ is the final semantic token with the same channel dimension with the input semantic tokens $\boldsymbol{ST}''$.

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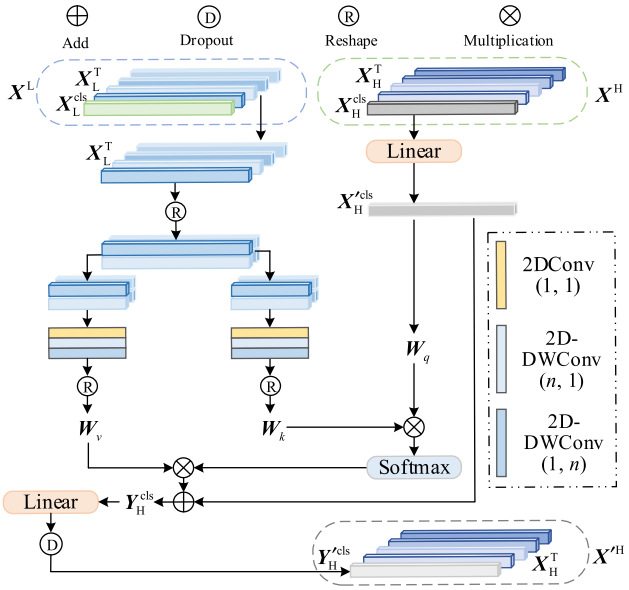


Fig. 4. Architecture of the proposed OCA. ($\boldsymbol{X}^{\mathrm{L}}$ represents the new total semantic tokens of LiDAR-DSM, where $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{T}}$ is the feature token and $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{cls}}$ is the class token. $\boldsymbol{X}^{\mathrm{H}}$ represents the new total semantic tokens of HSI, where $\boldsymbol{X}_{\mathrm{H}}^{\mathrm{T}}$ is the feature token and $\boldsymbol{X}_{\mathrm{H}}^{\mathrm{cls}}$ is the class token. $\boldsymbol{X}_{\mathrm{H}}^{\prime\mathrm{cls}}$ is the new class token of HSI after Linear(·). $\boldsymbol{X}^{\prime\mathrm{H}}$ is the new fusion semantic tokens of HSI, where $\boldsymbol{Y}_{\mathrm{H}}^{\mathrm{cls}}$ is the new fusion class token. $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$ are the learnable weights.)

Finally, execute the LN and residual connection again. The times of LGTM encoder loops can be set depending on the task and data.

C. OCA Fusion Encoder

To combine the advantage of CNN in extracting local information with the ability of the transformer to establish long-range dependencies, this article, inspired by the optimization-inspired cross-attention transformer (OCT) [56] and locally enhanced feed-forward network (LeFF) [57], proposes the fusion encoder OCA. In each encoder, the ability to capture global similarities between tokens is retained while enhancing the extraction of local information. Taking the OCA of HSI semantic tokens as an example in Fig. 4. The class token $\boldsymbol{X}_{\mathrm{H}}^{\mathrm{cls}}$ of HSI is through the Linear(·) first

$$\boldsymbol{X}_{\mathrm{H}}^{\prime\mathrm{cls}} = \mathrm{Linear}\left(\boldsymbol{X}_{\mathrm{H}}^{\mathrm{cls}}\right) \tag{10}$$

where $\boldsymbol{X}_{\mathrm{H}}^{\mathrm{cls}}$ is the class token from HSI features representations and has a size of $1 \times d_{\mathrm{H}}$. $\boldsymbol{X}_{\mathrm{H}}^{\prime\mathrm{cls}}$ represents the transformed class token, which dimension is consistent with $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{T}}$. Linear(·) is used to achieve dimensional alignment.

The feature tokens of LiDAR-DSM are retained as follows:

$$\boldsymbol{X}_{\mathrm{L}}^{\mathrm{T}} = \boldsymbol{X}^{\mathrm{L}} - \boldsymbol{X}_{\mathrm{L}}^{\mathrm{cls}} \tag{11}$$

where $\boldsymbol{X}^{\mathrm{L}}$ is the semantic token learned from LGTM of the LiDAR-DSM branch. $\boldsymbol{X}_{\mathrm{L}}{}^{\mathrm{cls}}$ is the class token of LiDAR-DSM. $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{T}}$ has a size of $N \times d_{\mathrm{L}}$. Then, OCA, a cross-attention (CA) mechanism that introduces the ADC, is used to fusion encode between $\boldsymbol{X}_{\mathrm{H}}^{\prime\mathrm{cls}}$ and $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{T}}$. $\boldsymbol{X}_{\mathrm{H}}^{\prime\mathrm{cls}}$ is the unique query vector for attention operations because the semantic tokens information of HSI are fused into the class token. $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{T}}$ is spatially reconstructed into "images" of $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{S}} \in R^{\sqrt{N}\times\sqrt{N}\times d_{\mathrm{L}}}$ based on its original locations. Then, $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{S}}$ is evenly divided into two parts of $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{S}_1} \in R^{\sqrt{N}\times\sqrt{N}\times(d_{\mathrm{L}}/2)}$ and $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{S}_2} \in R^{\sqrt{N}\times\sqrt{N}\times(d_{\mathrm{L}}/2)}$ in the channel dimension. Two consecutive ADC with $n \times 1$ and $1 \times n$ are performed on these two parts, whose outputs are

$$\boldsymbol{X}_{\mathrm{L}}^{\prime\mathrm{S}_1} = \mathrm{ADC}_{\text{2-D}}^{1\times n}\left(\mathrm{ADC}_{\text{2-D}}^{n\times 1}\left(C_{\text{2-D}}^{1\times 1}\left(\boldsymbol{X}_{\mathrm{L}}^{\mathrm{S}_1}\right)\right)\right) \tag{12}$$

$$\boldsymbol{X}_{\mathrm{L}}^{\prime\mathrm{S}_2} = \mathrm{ADC}_{\text{2-D}}^{1\times n}\left(\mathrm{ADC}_{\text{2-D}}^{n\times 1}\left(C_{\text{2-D}}^{1\times 1}\left(\boldsymbol{X}_{\mathrm{L}}^{\mathrm{S}_2}\right)\right)\right) \tag{13}$$

where $\boldsymbol{X}_{\mathrm{L}}^{\prime\mathrm{S}_1} \in R^{\sqrt{N}\times\sqrt{N}\times(h_n\times d_{\mathrm{L}})}$ and $\boldsymbol{X}_{\mathrm{L}}^{\prime\mathrm{S}_2} \in R^{\sqrt{N}\times\sqrt{N}\times(h_n\times d_{\mathrm{L}})}$. h_n is the number of attention heads and d_{L} is the channel dimension of features tokens for LiDAR-DSM. N is the number of $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{T}}$ and n is the convolutional kernel size.

The feature fusion based on OCA can be expressed as follows:

$$\begin{cases} \boldsymbol{Q} = \boldsymbol{X}_{\mathrm{H}}^{\prime\mathrm{cls}}\boldsymbol{W}_q \\ \boldsymbol{K} = \boldsymbol{X}_{\mathrm{L}}^{\prime S_1}\boldsymbol{W}_k \\ \boldsymbol{V} = \boldsymbol{X}_{\mathrm{L}}^{\prime S_2}\boldsymbol{W}_v \end{cases} \tag{14}$$

$$\boldsymbol{X}_{\mathrm{L}}^{\mathrm{H}} = \mathrm{OCA}\left(\boldsymbol{X}_{\mathrm{L}}^{\mathrm{T}}\right) = \mathrm{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{\mathrm{T}}}{\sqrt{D/h_n}}\right)\boldsymbol{V} \tag{15}$$

where $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$ are the weight matrices, D is the embedded dimension, and h_n is the number of attention heads. $\boldsymbol{X}_{\mathrm{L}}{}^{\mathrm{H}}$ contains the features information both HSI and LiDAR-DSM. OCA uses multiple heads, namely, multihead optimization convolution cross attention (MOCA).

Finally, the output $\boldsymbol{X}_{\mathrm{H}}^{\prime}$ of the OCA fusion encoder, which has LN and residual connection, is expressed as follows:

$$\boldsymbol{Y}_{\mathrm{H}}^{\mathrm{cls}} = \boldsymbol{X}_{\mathrm{H}}^{\prime\ \mathrm{cls}} + \mathrm{MOCA}\left(\mathrm{LN}\left(\boldsymbol{X}_{\mathrm{L}}^{\mathrm{H}}\right)\right) \tag{16}$$

$$\boldsymbol{Y}_{\mathrm{H}}^{\prime\ \mathrm{cls}} = \mathrm{Dropout}\left(\mathrm{Linear}\left(\boldsymbol{Y}_{\mathrm{H}}^{\mathrm{cls}}\right)\right) \tag{17}$$

$$\boldsymbol{X}^{\prime\mathrm{H}} = \boldsymbol{Y}_{\mathrm{H}}^{\prime\ \mathrm{cls}} \cup \left(\boldsymbol{X}^{\mathrm{H}} - \boldsymbol{X}_{\mathrm{H}}^{\mathrm{cls}}\right) \tag{18}$$

where $\boldsymbol{Y}_{\mathrm{H}}^{\mathrm{cls}}$ is the class token acquired by studying fusion features. It is consistent with the class token dimensions of $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{T}}$. $\boldsymbol{Y}_{\mathrm{H}}^{\prime\mathrm{cls}}$ indicates a class token with the same dimension as $\boldsymbol{X}_{\mathrm{H}}^{\mathrm{cls}}$, obtained by Linear(·). Linear(·) is used to achieve dimensional alignment.

The operations of the OCA encoder for the LiDAR-DSM branch are the same as that for the HSI branch. However, $\boldsymbol{Q}$ is from $\boldsymbol{X}_{\mathrm{L}}^{\mathrm{cls}}$, and $\boldsymbol{K}$ and $\boldsymbol{V}$ are from $\boldsymbol{X}_{\mathrm{H}}^{\mathrm{T}}$. The output after the fusion encoder is $\boldsymbol{X}^{\prime\mathrm{L}}$, and the class token is $\boldsymbol{Y}_{\mathrm{L}}^{\prime\mathrm{cls}}$.

D. Final Classifier

The overall process of CITNet is shown in Algorithm 1. Place class tokens $\boldsymbol{Y}_{\mathrm{H}}^{\prime\mathrm{cls}}$ and $\boldsymbol{Y}_{\mathrm{L}}^{\prime\mathrm{cls}}$ obtained from the OCA into the MLP to perform the final classification. Each MLP comprises two linear layers interspersed with a Gaussian error linear unit (GELU). The final linear layer equipped with a Softmax(·) is used to generate the ultimate labels for joint classification. The size of the output layer within the MLP is configured to the total number of classes, ensuring that the Softmax(·) normalizes the activation of output units and is summed up to 1. This normalization allows the output to represent a set of conditional probabilities. We obtain the conclusive probability vector by summing the final two output probability vectors, where the label with the highest probability denotes

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TABLE I
DETAILS ON THE TR DATASET, THE AU DATASET, THE MU DATASET, AND THE HU DATASET

No.	Color	TR			AU		
		Class Name	Training Samples	Test Samples	Class Name	Training Samples	Test Samples
1		Apple Trees	129	3905	Forest	675	12832
2		Buildings	125	2778	Residential Area	1516	28813
3		Ground	105	374	Industrial Area	192	3659
4		Woods	154	9896	Low Plants	1342	25515
5		Vineyard	184	10317	Allotment	28	547
6		Roads	122	3052	Commercial Area	82	1563
7					Water	76	1454
-	-	Total	819	29395	Total	3911	74383
No.	**Color**	**MU**			**HU**		
		Class Name	Training Samples	Test Samples	Class Name	Training Samples	Test Samples
1		Trees	150	23096	Healthy Grass	198	1053
2		Mostly Grass	150	4120	Stressed Grass	190	1064
3		Mixed Ground Surface	150	6732	Synthetic Grass	192	505
4		Dirt and Sand	150	1676	Trees	188	1056
5		Road	150	6537	Soil	186	1056
6		Water	150	316	Water	182	143
7		Buildings Shadow	150	2083	Residential	196	1072
8		Buildings	150	6090	Commercial	191	1053
9		Sidewalk	150	1235	Road	193	1059
10		Yellow Curb	150	33	Highway	191	1036
11		Cloth Panels	150	119	Railway	181	1054
12					Parking Lot 1	192	1041
13					Parking Lot 2	184	285
14					Tennis Court	181	247
15					Running Track	187	473
-	-	Total	1650	52037	Total	2832	12197

The choices of training samples and test samples are consistent with the main baseline method HCT [37].

the pixel category. Finally, the cross-entropy (CE) lossfunction is employed to calculate the loss.

IV. EXPERIMENTS AND RESULTS ANALYSIS

A. Datasets Description

The classification performance of CITNet is evaluated on four multimodal datasets: Trento (TR) [37], Augsburg (AU) [58], [59], MUUFL (MU) [60], [61], and Houston2013 (HU) [37]. Below are the details of all datasets. Table I shows the color, the class name, the training samples, and the test samples of four datasets.

1) TR dataset was collected around TR, Italy. It includes both HSI and LiDAR-DSM data. TR has a spatial size of 600 × 166 and six categories. HSI has 63 bands ranging from 420.89 to 989.09 nm, with a spectral resolution of 9.2 nm and a spatial resolution of 1 m. LiDAR-DSM is a single-channel image that contains the position height of the ground. Fig. 5 shows the false-color map of HSI, the grayscale map of LiDAR-DSM, and the ground-truth map.
2) The AU dataset was captured over AU, Germany, with a pixel size of 332 × 485 and seven categories. HSI was obtained by the DAS-EOC HySpex sensor, and the DDLR-3K system collected LiDAR-DSM. The spatial resolution of HSI and LiDAR-DSM was reduced to 30 m uniformly. HSI has 180 bands ranging from 0.4 to 2.5 μm, while LiDAR-DSM has only one raster. Fig. 6 shows the false-color map of HSI, the grayscale map of LiDAR-DSM, and the ground-truth map.
3) The MU dataset was collected in November 2010 at the Gulf Park Campus of the University of Southern Mississippi in Long Beach, Mississippi, USA. It has HSI and LiDAR-DSM data, with a 1-m spatial resolution and 11 categories. HSI imaging between 380 and 1050 nm. Removing the first and last four bands affected by noise and the uncovered area of the original map, the final 325 × 220 pixels and 64 bands were obtained. Fig. 7

Algorithm 1 CITNet for Hyperspectral and LiDAR Image Classification

Input: HSI: $\boldsymbol{X}_{\text{ori}}^{\text{H}} \in R_{H\times W\times B}$, LiDAR-DSM: $\boldsymbol{X}_{\text{ori}}^{\text{L}} \in R_{H\times W}$, labels: $\boldsymbol{Y}_{\text{L}} \in R_{H\times W}$, patches: 11 × 11, PCA: 30.
Output: Prediction: $\boldsymbol{Y}_{\text{P}}$.
Initialize: Batch size: 64, epochs: 100, the initial learning rate of the optimizer Adam depends on datasets.
PCA: $\boldsymbol{X}_{\text{PCA}}^{\text{H}} \in R_{H\times W\times C}$.
Patches: Accomplish the slicing process for 3D and 2D data to acquire the small patches $\boldsymbol{X}_{\text{in}}^{\text{H}} \in R_{P\times P\times C}$ and $\boldsymbol{X}_{\text{in}}^{\text{L}} \in R_{P\times P}$.
Split $\boldsymbol{X}_{\text{in}}^{\text{H}}$ and $\boldsymbol{X}_{\text{in}}^{\text{L}}$ into training sets $\boldsymbol{D}_{\text{train}}$ and test sets $\boldsymbol{D}_{\text{test}}$ ($\boldsymbol{D}_{\text{train}}$ contains the labels, and $\boldsymbol{D}_{\text{test}}$ does not contain the labels).
Training CITNet (begin):
for epoch in range (epochs):
  for i, ($\boldsymbol{D}_{\text{train}}^{\text{H}}$, $\boldsymbol{D}_{\text{train}}^{\text{L}}$, $\boldsymbol{Y}_{\text{L}}$) in enumerate ($\boldsymbol{D}_{\text{train}}$):
    Generate shallow local-global features using the **MADC**.
    Perform the LGTM module:
    Perform tokenization transformation using the 1D feature vectors derived from MADC to initialize weights and generate semantic tokens by Eq. (1). A learnable class token T_0^{cls} is applied to the first position of the semantic tokens, while the positional embedding is carried out to the semantic tokens. Learn the new semantic tokens by Eq. (2) and (3), and perform LGF by Eq. (4) to Eq. (9).
    Input the learned semantic tokens to the OCA encoder for acquiring the fusion features.
    Perform OCA fusion encoder by Eq. (10) to Eq. (18).
    Input the semantic tokens from the OCA fusion encoder into the MLP: $\boldsymbol{X}_{\text{out}} = \text{MLP}(\boldsymbol{X}_{\text{OCA}}^{\text{H}}) \oplus \text{MLP}(\boldsymbol{X}_{\text{OCA}}^{\text{L}})$.
    $\text{CE}(\boldsymbol{X}_{\text{out}}, \boldsymbol{Y}_{\text{L}})$.
  end for
end for
Training CITNet (end) and test CITNet:
$\boldsymbol{Y}_{\text{P}} = \text{CITNet}_{\text{trained}}(\boldsymbol{D}_{\text{test}})$.

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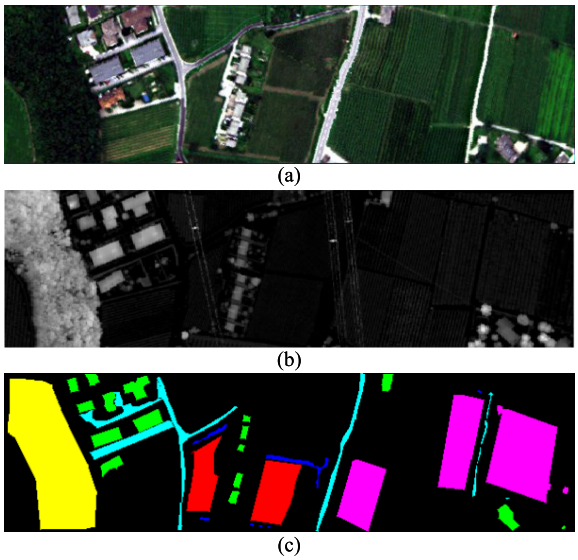


Fig. 5. TR dataset. (a) Pseudocolor map. (b) Grayscale map. (c) Ground-truth map.

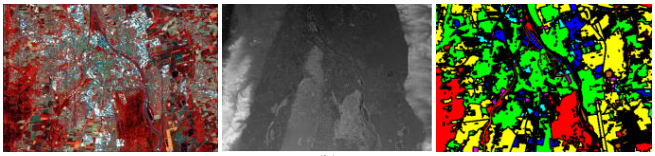


Fig. 6. AU dataset. (a) Pseudocolor map. (b) Grayscale map. (c) Ground-truth map.

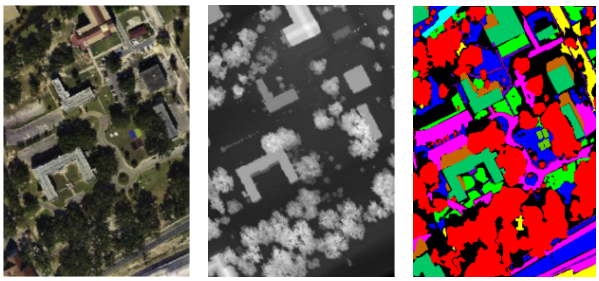


Fig. 7. MU dataset. (a) Pseudocolor map. (b) Grayscale map. (c) Ground-truth map.

shows the false-color map of HSI, the grayscale map of LiDAR-DSM, and the ground-truth map.

4) The HU dataset was derived from the 2013 IEEE GRSS Data Fusion Competition. The scene captures the University of Houston and its surrounding area in TX, USA. It includes HSI and LiDAR-DSM data of 340 × 1905 pixels with a spatial resolution of 2.5 m across 15 categories. HSI has 144 bands with a wavelength range of 0.38– 1.05 μm. Fig. 8 shows the false-color map of HSI, the grayscale map of LiDAR-DSM, and the ground-truth map.

B. Experimental Settings

Experiments are performed on Windows 11, Intel Core i9 CPU with 32-GB memory, and RTX 3090Ti graphics with 24-GB memory, coding by Python 3.8 under Pytorch 1.12.0. The input patch size, epochs, and batch size are 11 × 11, 100, and 64. The number of spectra chosen by PCA is 30. The optimization algorithm is Adam, whose initial learning rates for TR, AU, MU, and HU datasets are 0.0005, 0.001, and 0.0005, respectively. The dropout rates in MACT are set to 0.5 for all datasets. The number of tokens after tokenization for all datasets is 4, the same as the baseline. The convolutional kernel size in OCA is 5. Training and testing samples are selected randomly to improve the reliability of the experiments. Selections of training and testing samples of all datasets as shown in Table I, the same as the primary baseline method HCT [37], to ensure the equity of comparison experiments. All experiments are conducted five consecutive times and averaged. Overall accuracy (OA), average accuracy (AA), and statistical kappa coefficient (K) are the objective evaluation indicators.

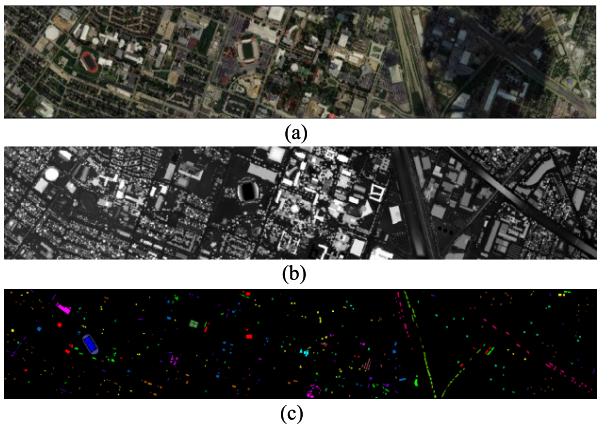


Fig. 8. HU dataset. (a) Pseudocolor map. (b) Grayscale map. (c) Ground-truth map.

Comparisons of different experimental parameters are conducted to determine the optimal model configuration. The initial learning rates of Adam, the attention heads and the depth of the encoder in LGTM, and the attention heads and the depth of OCA are tested on four datasets. To control the variables, the input size, the epoch, the number of experiments, and the number of training and testing samples are consistent.

1) *Initial Learning Rates of the Adam:* Fig. 9 shows the results of different initial learning rates (i.e., 0.001, 0.0005, and 0.0001). The analysis reveals that optimal classification results are achieved by setting the initial learning rate to 0.0005 for the TR, AU, and HU datasets. In contrast, the MU dataset benefits most from a learning rate of 0.001. These findings underscore the importance of fine-tuning learning rates to maximize model accuracy across varying datasets.
2) *Selections of the Patch Size:* Fig. 10 shows the results of different patch sizes. The five patch sizes (i.e., 7 × 7, 9 × 9, 11 × 11, 13 × 13, and 15 × 15) are selected for experiments. To the TR, MU, and HU datasets, the OA, AA, and K × 100 are highest when the patch size is 11 × 11. To the AU dataset, OA and K K × 100 are similar in all patch sizes, and AA is higher when the patch size is 15 × 15, which is a little higher than 11 × 11. The results show that when the patch size is 11 × 11 simultaneously, the comprehensive results of the four datasets are optimal.

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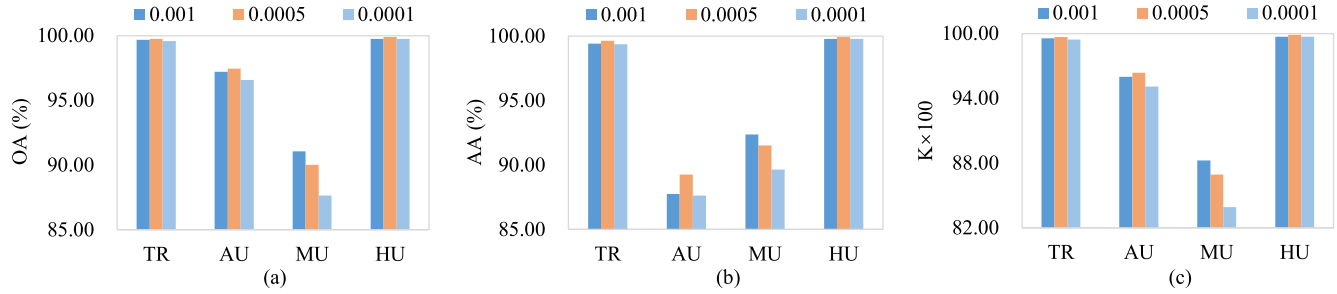


Fig. 9. Choice of initial learning rate. (a) OA (%), (b) AA (%), and (c) K × 100. The horizontal coordinate represents different datasets.

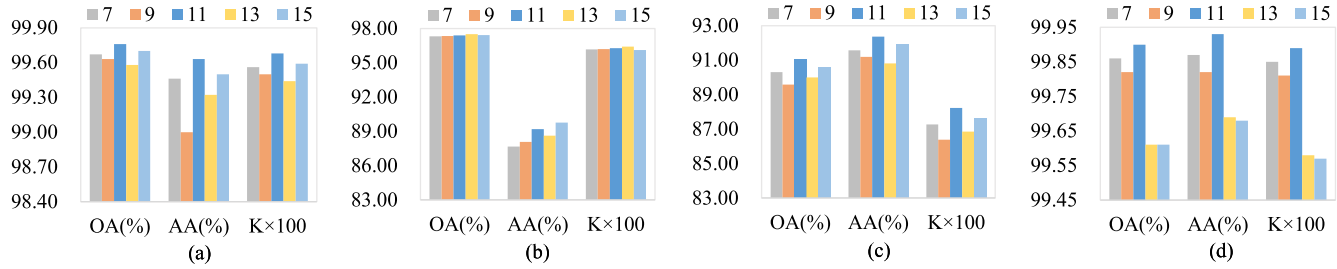


Fig. 10. Effects of different patch sizes. (a) TR dataset, (b) AU dataset, (c) MU dataset, and (c) HU dataset. The ordinate represents the accuracy.

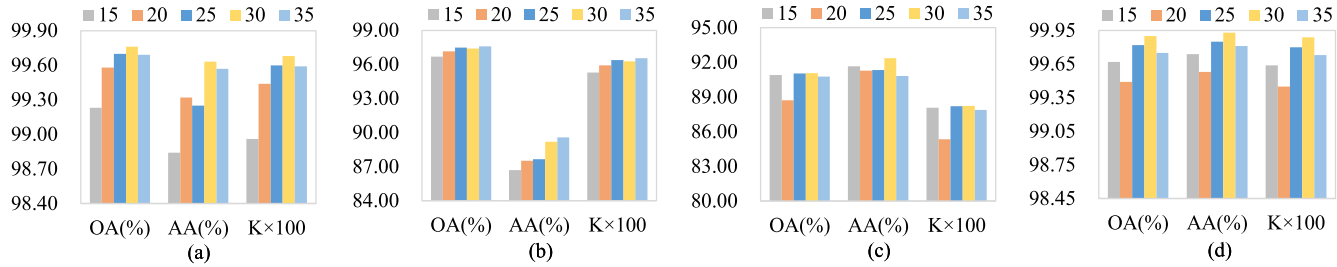


Fig. 11. Effects of different HSI bands. (a) TR dataset, (b) AU dataset, (c) MU dataset, and (c) HU dataset. The ordinate represents the accuracy.

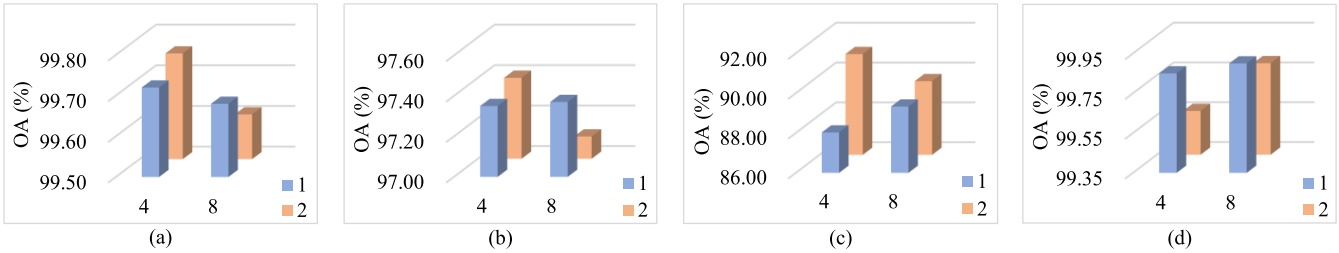


Fig. 12. Joint effects of the CIT block depth, attention heads in LGTM, and the encoder depth of LGTM. (a) TR dataset (4 + 2). (b) AU dataset (4 + 2). (c) MU dataset (4 + 2). (d) HU dataset (8 + 1). 4 and 8 are attention heads. Blue represents one encoder depth and one CIT block depth, and orange represents two.

3) *Selections of HSI Bands:* Fig. 11 shows the classification results of different numbers of bands selected by PCA, such as 15, 20, 25, 30, and 35. To the TR and HU datasets, the OA, AA, and K × 100 are highest when HSI bands after PCA are 30. To the AU dataset, OA, AA, and K are highest when 35 HSI bands, a little over 30 HSI bands. To the MU dataset, when HSI bands are 25 and 30, OA and K are similar and higher than other cases, and AA is much higher when HSI bands are 30. Verified by experiments, input bands of HSI are chosen 30 to generalize the proposed method on four datasets.
4) *CIT Block Depth, and Attention Heads and Encoder Depth in LGTM:* Fig. 12 illustrates the evaluation of four different configurations of attention heads and encoder depths (i.e., 4 + 1, 4 + 2, 8 + 1, and 8 + 2). The results indicate that the optimal classification performance is achieved with a 4 + 2 configuration for the TR, MU, and AU datasets. The HU dataset achieves the best results with an 8 + 1 configuration.
5) *Attention Heads and Encoder Depth of OCA:* Fig. 13 shows four combinations of attention heads and the encoder depth of OCA (i.e., 4 + 1, 4 + 2, 8 + 1, and 8 + 2). The results show that the TR dataset is 4 + 2, the MU dataset is 8 + 2, and the HU dataset is 8 + 1 to obtain optimal classification capability. 4 + 1 and 4 + 2 have similar classification accuracy on AU datasets, while 4 + 1 produces lower model parameters, so 4 + 1 is chosen.

C. Experimental Results and Analysis

To verify the classification performance, the proposed CITNet is compared with LiEtAl [62], SpectralFormer [24],

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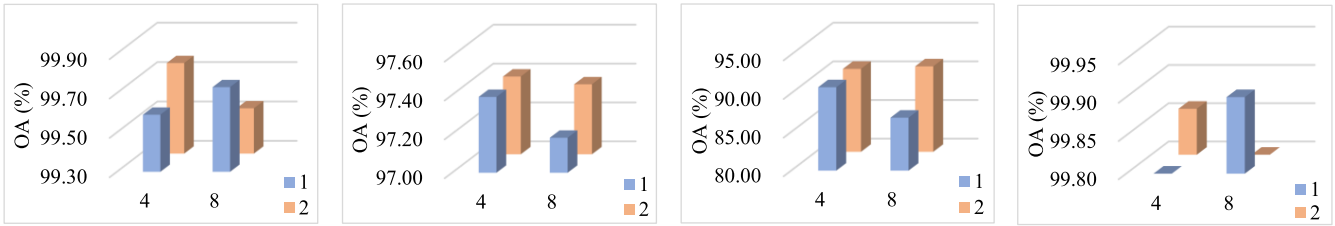


Fig. 13. Joint effects of attention heads and the encoder depth in OCA. (a) TR dataset (4 + 2). (b) AU dataset (4 + 1). (c) MU dataset (8 + 2). (d) HU dataset (8 + 1). 4 and 8 are attention heads. Blue represents one encoder depth and orange represents two.

TABLE II

CLASSIFICATION RESULTS OF ALL METHODS ON TR DATASET

NO.	HSI Input				HSI and LiDAR-DSM Input					
	LiEtAl	SpectralFormer	SSFTT	morpFormer	MDL_Cros	MFT_PT	MFT_CT	HCT	TMCFN	**CITNet**
1	99.39 ± 0.42	99.51 ± 0.21	99.84 ± 0.05	98.26 ± 0.42	96.15 ± 5.47	97.15 ± 0.09	97.72 ± 0.91	99.64 ± 0.18	**99.87 ± 0.02**	99.63 ± 0.32
2	92.61 ± 1.47	94.92 ± 2.50	97.47 ± 0.55	98.52 ± 0.60	98.63 ± 0.83	99.44 ± 0.23	97.62 ±0.64	98.83 ± 0.49	97.20 ± 0.29	**99.67 ± 0.30**
3	97.75 ± 0.37	98.04 ± 0.77	99.84 ± 0.21	**100 ± 0.00**	96.81 ± 3.19	99.37 ± 0.70	99.82 ± 0.12	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**
4	99.88 ± 0.12	99.98 ± 0.02	**100 ± 0.00**	**100 ± 0.00**	99.97 ± 0.03	99.99 ± 0.01	99.99 ± 0.01	**100 ± 0.00**	99.90 ± 0.01	**100 ± 0.00**
5	99.62 ± 0.40	99.59 ± 0.29	**99.99 ± 0.01**	99.91 ± 0.05	97.03 ± 1.34	99.96 ± 0.02	99.94 ± 0.01	99.96 ± 0.06	99.20 ± 0.04	99.97 ± 0.03
6	91.16 ± 2.99	89.67 ± 4.70	95.00 ± 0.91	96.89 ± 0.72	98.82 ± 0.65	94.47 ± 1.08	98.75 ± 0.17	97.50 ± 0.83	**99.89 ± 0.05**	98.54 ± 0.57
OA (%)	98.10 ± 0.24	98.21 ± 0.21	99.22 ± 0.07	99.27 ± 0.04	98.15 ± 0.46	98.97 ± 0.13	99.32 ± 0.08	99.56 ± 0.09	99.39 ± 0.02	**99.76 ± 0.04**
AA (%)	96.73 ± 0.41	96.95 ± 0.47	98.69 ± 0.11	98.93 ± 0.06	97.90 ± 1.10	98.40 ± 0.21	98.97 ± 0.11	99.32 ± 0.13	99.34 ± 0.05	**99.63 ± 0.05**
K × 100	97.46 ± 0.26	97.60 ± 0.29	98.95 ± 0.09	99.03 ± 0.05	97.53 ± 0.65	98.62 ± 0.18	99.09 ± 0.11	99.42 ± 0.12	99.19 ± 0.03	**99.68 ± 0.05**

The bold represents the best accuracy, and the underline represents the second-best accuracy.

TABLE III

CLASSIFICATION RESULTS OF ALL METHODS ON AU DATASET

NO.	HSI Input				HSI and LiDAR-DSM Input					
	LiEtAl	SpectralFormer	SSFTT	morpFormer	MDL_Cross	MFT_PT	MFT_CT	HCT	TMCFN	**CITNet**
1	98.25 ± 0.43	98.80 ± 6.67	99.19 ± 0.24	98.71 ± 0.35	97.85 ± 0.76	97.52 ± 0.12	97.24 ± 0.54	99.03 ± 0.31	98.96 ± 0.09	**99.36 ± 0.13**
2	77.97 ± 3.56	96.28 ± 1.11	**99.04 ± 0.17**	98.42 ± 0.45	83.37 ± 5.41	98.53 ± 0.38	98.44 ± 0.16	98.70 ± 0.36	91.65 ± 1.29	98.74 ± 0.43
3	**97.54 ± 0.48**	75.47 ± 0.22	82.53 ± 0.22	89.67 ± 1.24	90.01 ± 7.72	84.69 ± 0.46	87.21 ± 0.41	87.24 ± 2.28	94.30 ± 0.29	90.52 ± 3.63
4	98.54 ± 0.22	98.77 ± 5.31	98.70 ± 0.21	98.00 ± 0.17	**99.56 ± 0.25**	97.91 ± 0.09	97.39 ± 0.67	98.73 ± 0.52	97.93 ± 0.73	98.89 ± 0.24
5	64.97 ± 5.04	46.80 ± 0.44	88.84 ± 3.08	80.37 ± 3.78	19.82 ± 8.86	79.09 ± 5.14	75.92 ± 2.12	84.69 ± 3.61	**99.83 ± 0.01**	91.15 ± 3.18
6	46.60 ± 2.01	39.15 ± 7.52	80.97 ± 1.97	74.60 ± 0.53	44.84 ± 8.44	77.86 ± 3.07	71.46 ± 4.91	69.29 ± 2.72	**92.59 ± 0.49**	76.44 ± 6.90
7	58.39 ± 3.58	59.30 ± 2.66	68.45 ± 1.60	65.86 ± 1.33	60.48 ± 5.33	70.77 ± 4.13	69.73 ± 1.18	71.27 ± 7.74	**92.88 ± 0.92**	69.24 ± 1.50
OA (%)	94.84 ± 0.39	94.26 ± 0.47	97.08 ± 0.14	96.63 ± 0.19	90.03 ± 8.85	96.34 ± 0.10	96.03 ± 0.30	96.89 ± 0.21	95.29 ± 0.64	**97.40 ± 0.10**
AA (%)	71.66 ± 0.85	73.51 ± 1.70	88.25 ± 0.39	86.52 ± 0.61	70.85 ± 2.82	86.62 ± 1.60	85.34 ± 0.70	86.56 ± 1.05	**95.45 ± 0.15**	89.19 ± 0.76
K × 100	92.58 ± 0.79	91.76 ± 0.68	95.82 ± 0.20	95.18 ± 0.27	85.94 ± 5.73	94.77 ± 0.14	94.32 ± 0.43	95.53 ± 0.34	93.37 ± 0.30	**96.27 ± 0.14**

The bold represents the best accuracy, and the underline represents the second-best accuracy.

SSFTT [25], morpFormer [63], MDL_Cross [64], MFT_PT [36], MFT_CT [36], HCT [37], and TMCFN [65]. Hyperparameter settings of morpFormer, MFT_PT, MFT_CT, TMCFN, and baseline HCT align with their studies. According to the literature provided by TMCFN, the epoch is 1000 for optimal performance. The initial learning rates of LiEtAl, SpectralFormer, SSFTT, and MDL_Cross are the same as CITNet to get the best outcome. The classification results from Tables II to V show that CITNet achieves the best comprehensive classification effect on four datasets. From Figs. 14 to 17, the higher the classification accuracy, the salt-and-pepper noise in the corresponding classification maps decreases. The proposed CITNet has the fewest salt-and-pepper noise.

1) *TR:* From Table II, LiEtAl has the worst classification results, and MDL_Cross is the second worst. The proposed CITNet improves 1.66%, 1.55%, 0.54%, 0.49%, 1.61%, 0.79%, 0.44%, 0.20%, and 0.37% on OA compared with LiEtAl, SpectralFormer, SSFTT, morpFormer, MDL_Cross, MFT_PT, MFT_CT, HCT, and TMCFN. At the same time, the CITNet improves 2.90%, 2.68%, 0.94%, 0.70%, 1.73%, 1.23%, 0.66%, 0.31%, and 0.29% on AA, and improves 2.22, 2.08, 0.73, 0.65, 2.15, 1.06, 0.59, 0.26, and 0.49 on K × 100, respectively. In addition, the accuracies of categories 3 and 4 of SSFTT, morpFormer, HCT, TMCFN, and the proposed CITNet reached 100%. Because the total number of the ground is few and its training samples are relatively large, the features learning is adequate. However, although the training samples of the woods are few, the sample distribution is simple, which is convenient for feature extraction. Fig. 14 reveals that the classification map generated by CITNet exhibits the closest resemblance to the ground-truth map among the evaluated methods.
2) *AU:* From Table III, MDL_Cross has the worst classification results, and SpectralFormer is the second worst. OA of CITNet increased by 2.56%, 3.14%, 0.32%, 0.77%, 7.37%, 1.06%, 1.37%, 0.51%, and 2.11% compared with LiEtAl, SpectralFormer, SSFTT, morpFormer, MDL_Cross, MFT_PT, MFT_CT, HCT, and TMCFN. Simultaneously, AA of CITNet increased by 17.53%, 15.68%, 0.94%, 2.67%, 18.34%, 2.57%, 3.85%, 2.63%, and −6.26%, and K × 100 increased by 3.69,

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TABLE IV
CLASSIFICATION RESULTS OF ALL METHODS ON MU DATASET

NO.	HSI Input				HSI and LiDAR-DSM Input					
	LiEtAl	SpectralFormer	SSFTT	morpFormer	MDL_Cross	MFT_PT	MFT_CT	HCT	TMCFN	CITNet
1	85.06 ± 2.01	86.04 ± 0.98	88.43 ± 0.30	87.32 ± 1.55	80.34 ± 6.18	86.04 ± 2.52	85.48 ± 1.43	89.41 ± 1.77	88.85 ± 0.01	**93.17 ± 1.16**
2	77.97 ± 3.56	85.55 ± 2.13	83.68 ± 1.09	82.63 ± 6.05	71.89 ± 3.25	80.49 ± 3.43	81.98 ± 3.23	86.29 ± 0.17	**86.91 ± 1.92**	86.29 ± 4.04
3	74.57 ± 5.76	76.98 ± 7.22	82.86 ± 0.59	79.61 ± 4.96	62.09 ± 8.90	74.63 ± 7.14	75.06 ± 3.69	76.87 ± 0.81	83.98 ± 0.66	**84.51 ± 4.15**
4	89.03 ± 4.60	95.34 ± 0.85	95.80 ± 0.46	**97.63 ± 0.63**	97.46 ± 2.65	94.45 ± 1.13	93.40 ± 0.73	96.40 ± 1.11	95.13 ± 0.11	93.42 ± 2.39
5	81.11 ± 3.52	87.13 ± 1.36	84.18 ± 1.29	77.46 ± 2.77	73.17 ± 1.48	78.41 ± 5.31	78.40 ± 0.84	89.38 ± 1.69	87.53 ± 1.54	**89.63 ± 1.26**
6	99.49 ± 0.51	99.15 ± 0.53	98.83 ± 0.39	98.94 ± 0.83	99.81 ± 0.19	**99.89 ± 0.14**	98.92 ± 0.32	99.78 ± 0.14	99.79 ± 0.01	99.49 ± 0.42
7	87.48 ± 1.81	92.38 ± 1.01	89.58 ± 0.66	89.47 ± 1.82	93.49 ± 3.73	93.23 ± 1.45	88.83 ± 1.47	93.67 ± 2.04	96.53 ± 0.25	**96.54 ± 1.20**
8	88.90 ± 4.96	94.17 ± 1.09	90.66 ± 0.81	91.78 ± 2.88	94.53 ± 3.99	89.06 ± 2.65	95.01 ± 1.65	95.08 ± 1.03	**95.79 ± 0.33**	93.67 ± 2.19
9	64.96 ± 1.89	76.95 ± 2.44	78.54 ± 1.45	74.57 ± 4.62	43.45 ± 7.01	69.90 ± 4.66	69.65 ± 3.55	71.25 ± 4.13	**84.73 ± 0.47**	82.31 ± 3.69
10	92.75 ± 4.21	98.98 ± 1.42	92.92 ± 3.77	87.87 ± 2.47	76.90 ± 7.20	93.93 ± 2.47	95.15 ± 2.42	92.92 ± 1.42	**99.45 ± 0.01**	98.78 ± 1.48
11	97.14 ± 1.34	99.15 ± 0.52	**99.43 ± 0.39**	97.47 ± 2.05	97.48 ± 1.68	97.75 ± 0.39	99.32 ± 0.82	98.87 ± 0.39	99.26 ± 0.01	98.31 ± 1.71
OA (%)	82.88 ± 2.30	86.37 ± 0.59	89.54 ± 0.43	85.45 ± 0.93	78.35 ± 4.84	83.81 ± 0.92	84.21 ± 0.44	88.25 ± 0.74	89.32 ± 0.21	**91.06 ± 0.62**
AA (%)	85.13 ± 1.20	90.16 ± 0.23	89.54 ± 0.43	87.71 ± 0.17	80.97 ± 3.90	87.07 ± 0.15	87.38 ± 0.51	89.99 ± 0.49	**92.54 ± 0.14**	92.37 ± 0.21
K × 100	77.86 ± 0.93	82.35 ± 0.75	83.36 ± 0.21	81.14 ± 1.09	72.52 ± 5.63	79.08 ± 1.08	79.64 ± 0.54	84.68 ± 0.91	86.24 ± 0.27	**88.23 ± 0.78**

The bold represents the best accuracy, and the underline represents the second-best accuracy.

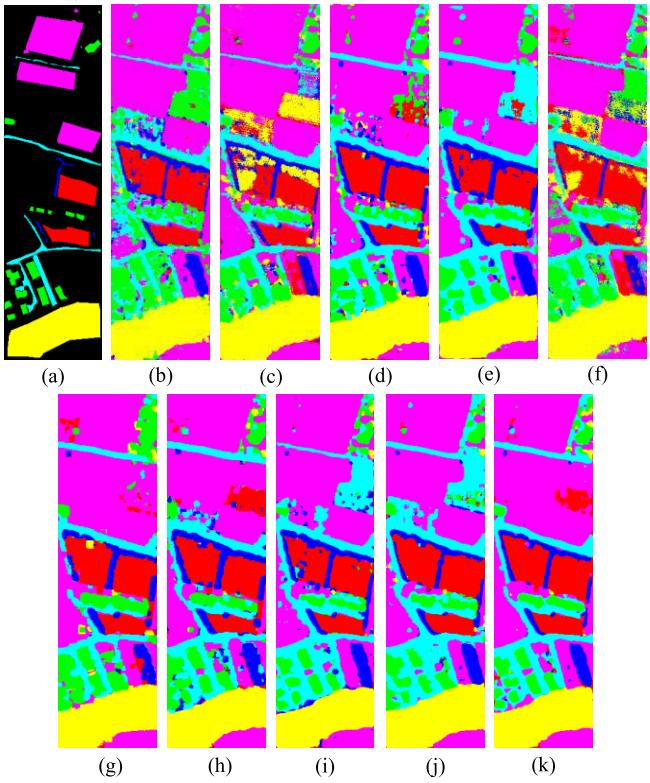


Fig. 14. Classification maps of different methods on TR dataset. (a) Ground-truth map. (b) LiEtAl (98.10%). (c) SpectralFormer (98.21%). (d) SSFTT (99.22%). (e) morpFormer (99.27%). (f) MDL_Cross (98.15%). (g) MFT_PT (98.97%). (h) MFT_CT (99.32%). (i) HCT (99.56%). (j) TMCFN (99.39%). (k) CITNet (proposed, 99.76%).

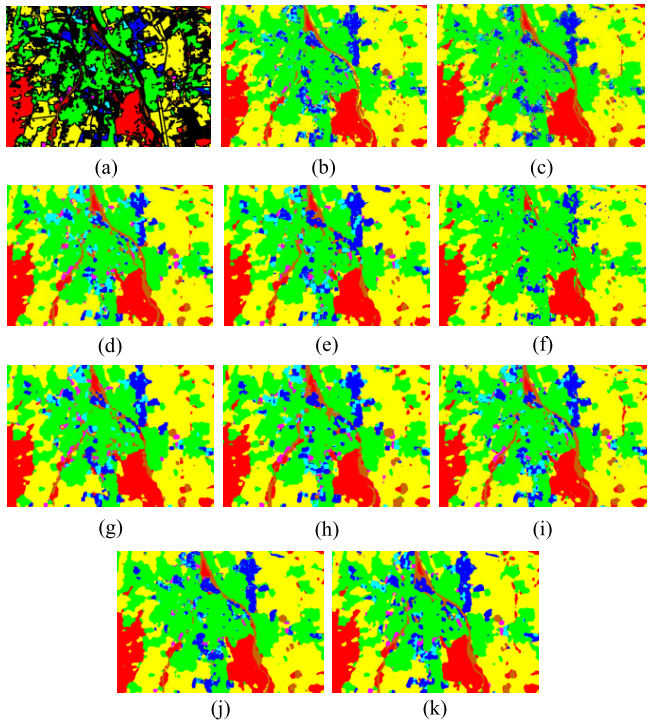


Fig. 15. Classification maps of different methods on AU dataset. (a) Ground-truth map. (b) LiEtAl (94.84%). (c) SpectralFormer (94.26%). (d) SSFTT (97.08%). (e) morpFormer (96.63%). (f) MDL_Cross (90.03%). (g) MFT_PT (96.34%). (h) MFT_CT (96.03%). (i) HCT (96.89%). (j) TMCFN (95.29%). (k) CITNet (proposed, 97.40%).

4.51, 0.45, 1.09, 10.33, 1.50, 1.95, 0.74, and 2.90, respectively. Sample distributions of Category 6 commercial areas and Category 7 water are complex, making it difficult to extract and learn features, and the classification results of most methods are low. TMCFN has text supplements to the features to get high classification performance. At the same time, CITNet exhibits minimal salt-and-pepper noise contrasted with other methods in Fig. 15.

3) *MU:* From Table IV, MDL_Cross has the worst classification results, and LiEtAl is the second worst. OA of the CITNet rise by 8.18%, 4.69%, 1.52%, 5.61%, 12.71%, 7.25%, 4.91%, 2.81%, and 1.74% compared with LiEtAl, SpectralFormer, SSFTT, morpFormer, MDL_Cross, MFT_PT, MFT_CT, HCT, and TMCFN. Meanwhile, the AA rise by 7.24%, 2.21%, 2.83%, 4.66%, 11.40%, 5.30%, 6.22%, 2.38%, and −0.17% and K × 100 rise by 10.37, 5.88, 4.87, 7.09, 15.71, 9.15, 6.29, 3.55, and 1.99, respectively. The heterogeneous and intricate sample distribution within the MU dataset poses a considerable challenge to classification accuracy using different methods. TMCFN has the feature description text, which can get a higher AA. CITNet captures a wealth of local–global feature information, which allows it to adeptly navigate the complexities of the sample distribution on the MU

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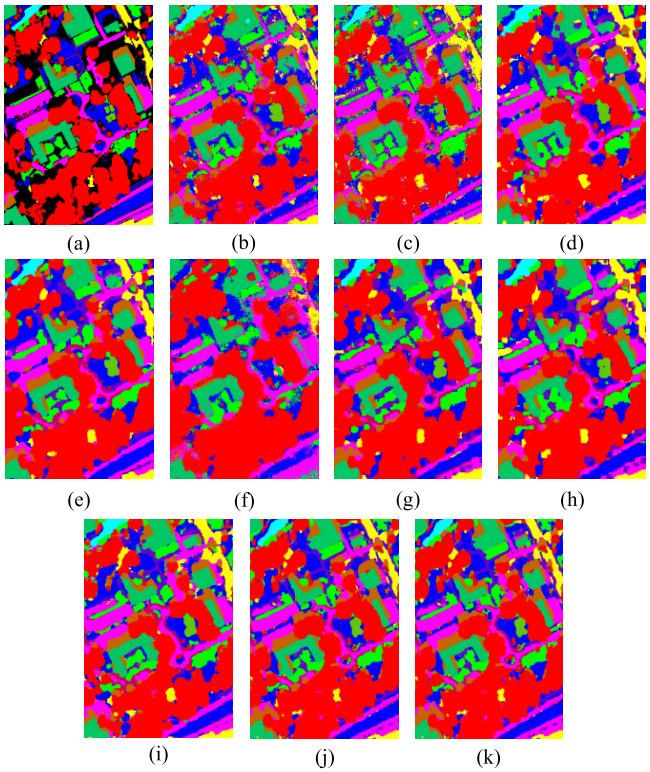


Fig. 16. Classification maps of different methods on MU dataset. (a) Ground-truth map. (b) LiEtAl (82.88%). (c) SpectralFormer (86.37%). (d) SSFTT (89.54%). (e) morpFormer (85.45%). (f) MDL_Cross (78.35%). (g) MFT_PT (83.81%). (h) MFT_CT (84.21%). (i) HCT (88.25%). (j) TMCFN (89.32%). (k) CITNet (proposed, 91.06%).

dataset, achieving a classification impact that surpasses that of competing algorithms. As illustrated in Fig. 16, the classification map produced by CITNet most closely aligns with the ground-truth map.

4) *HU:* From Table V, MDL_Cross has the worst classification results, and LiEtAl is the second worst. The classification accuracy of SSFTT is similar to that of MFT_CT. The proposed CITNet increased by 1.06%, 0.96%, 0.16%, 0.65%, 6.11%, 0.50%, 0.14%, 0.21%, and 0.27% on OA compared with LiEtAl, SpectralFormer, SSFTT, morpFormer, MDL_Cross, MFT_PT, MFT_CT, HCT, and TMCFN. The value of AA increased by 0.88%, 0.99%, 0.19%, 0.59%, 5.71%, 0.40%, 0.14%, 0.17%, and 0.23%, and the value of $K \times 100$ increased by 1.15, 1.04, 0.17, 0.71, 6.63, 0.54, 0.15, 0.23, and 0.29, respectively. SSFTT and HCT have similar classification performance. Due to the large proportion of training samples in Class 14 tennis court and Class 15 tennis court and the simple distribution feature, learning is sufficient. Therefore, most methods can achieve 100% accuracy. Fig. 17 illustrates that higher classification accuracy correlates with reduced salt-and-pepper noise, suggesting that CITNet effectively improves joint classification performance.

D. Consumption and Computational Complexity

In this section, we compare the training time (Train), test time (Test), total parameters (Params), and floating point operations (FLOPs) of all methods under a batch size of

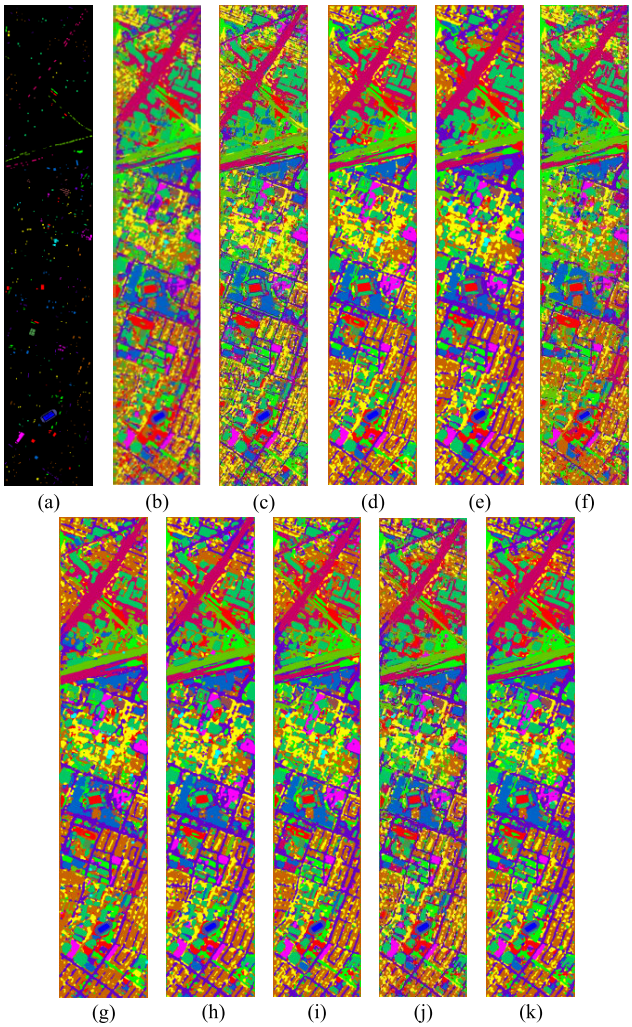


Fig. 17. Classification maps of different methods on HU dataset. (a) Ground-truth map. (b) LiEtAl (98.84%). (c) SpectralFormer (98.94%). (d) SSFTT (99.74%). (e) morpFormer (99.25%). (f) MDL_Cross (93.79%). (g) MFT_PT (99.40%). (h) MFT_CT (99.76%). (i) HCT (99.69%). (j) TMCFN (99.63%). (k) CITNet (proposed, 99.90%).

64 and 30 PCA components, as summarized in Table VI. The experimental settings remain consistent with those described previously. Notably, the depth of the encoder and the number of attention heads influence the total parameters and FLOPs of the proposed CITNet. Among the compared methods, TMCFN exhibits the highest total parameters and FLOPs on the TR, AU, and HU datasets, while CITNet leads in these metrics on the MU dataset. In addition, TMCFN demonstrates the longest training time on the TR, MU, and HU datasets, whereas CITNet requires the most extended training time on the AU dataset. CITNet surpasses all other comparison algorithms on test time. Despite this, the exceptionally high classification accuracy achieved by CITNet justifies the minor trade-offs in other performance aspects. Overall, CITNet delivers optimal performance across the board, highlighting its effectiveness and robustness in complex classification tasks.

V. Discussion

A. Ablation Analysis

1) *Different Components:* From Table VII, the first combination is the baseline algorithm HCT, which has the

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TABLE V
CLASSIFICATION RESULTS OF ALL METHODS ON HU DATASET

NO.	HSI Input				HSI and LiDAR-DSM Input					
	LiEtAl	SpectralFormer	SSFTT	morpFormer	MDL_Cross	MFT_PT	MFT_CT	HCT	TMCFN	**CITNet**
1	99.32 ± 0.17	98.82 ± 0.83	**99.90 ± 0.07**	99.84 ± 0.16	92.71 ± 5.48	98.10 ± 1.63	99.68 ± 0.19	99.62 ± 0.26	99.16 ± 0.20	99.66 ± 0.68
2	99.00 ± 0.25	99.96 ± 0.04	99.90 ± 0.07	98.27 ± 0.77	99.61 ± 0.55	99.34 ± 0.35	99.59 ± 0.34	99.96 ± 0.04	**100 ± 0.00**	99.76 ± 0.14
3	99.80 ± 0.20	99.93 ± 0.09	**100 ± 0.00**	98.48 ± 0.02	73.66 ± 7.33	99.93 ± 0.09	99.73 ± 0.18	**100 ± 0.00**	99.93 ± 0.07	**100 ± 0.00**
4	99.39 ± 0.72	99.74 ± 0.23	**99.96 ± 0.04**	99.33 ± 0.14	98.90 ± 0.81	99.58 ± 0.39	99.74 ± 0.22	99.51 ± 0.40	99.72 ± 0.20	99.79 ± 0.07
5	99.81 ± 0.19	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	99.17 ± 0.83	99.96 ± 0.04	**100 ± 0.00**	99.93 ± 0.04	**100 ± 0.00**	99.98 ± 0.04
6	99.44 ± 0.56	99.76 ± 0.35	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**
7	96.92 ± 1.77	96.73 ± 1.32	**100 ± 0.00**	98.56 ± 0.35	99.81 ± 0.10	99.68 ± 0.21	99.47 ± 0.68	99.93 ± 0.08	99.92 ± 0.08	99.98 ± 0.04
8	97.99 ± 1.34	98.32 ± 0.82	99.11 ± 0.04	98.51 ± 0.89	99.12 ± 0.87	**99.96 ± 0.04**	99.93± 0.08	99.43 ± 0.40	99.24 ± 0.12	99.96 ± 0.05
9	96.88 ± 1.60	97.38 ± 1.49	99.21 ± 0.77	98.55 ± 0.59	99.81 ± 0.19	98.77 ± 0.65	**99.93 ± 0.08**	98.55 ± 0.04	99.76 ± 0.01	99.81 ± 0.33
10	99.09 ± 0.35	99.42 ± 0.81	**100 ± 0.00**	99.93 ± 0.04	85.56 ± 14.4	**100 ± 0.00**	99.51 ± 0.68	**100 ± 0.00**	99.96 ± 0.04	**100 ± 0.00**
11	99.38 ± 0.24	99.55 ± 0.34	99.96 ± 0.21	99.93 ± 0.09	98.98 ± 1.02	99.96 ± 0.04	**100 ± 0.00**	99.84 ± 0.11	99.76 ± 0.16	99.98 ± 0.04
12	99.15 ± 0.27	99.19 ± 0.38	98.97 ± 0.37	99.35 ± 0.46	66.30 ± 1.66	97.82 ± 0.23	99.55 ± 0.23	99.61 ± 0.20	98.06 ± 0.17	**99.98 ± 0.04**
13	99.79 ± 0.21	95.32 ± 3.41	99.88 ± 0.07	99.29 ± 0.49	99.44 ± 0.56	99.88 ± 0.16	99.76 ± 0.33	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**
14	99.84 ± 0.16	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**
15	99.92 ± 0.08	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**	**100 ± 0.00**
OA (%)	98.84 ± 0.24	98.94 ± 0.37	99.74 ± 0.05	99.25 ± 0.20	93.79 ± 6.19	99.40 ± 0.08	99.76 ± 0.15	99.69 ± 0.03	99.63 ± 0.04	**99.90 ± 0.07**
AA (%)	99.05 ± 0.38	98.94 ± 0.47	99.79 ± 0.05	99.34 ± 0.21	94.22 ± 4.18	99.53 ± 0.05	99.79 ± 0.14	99.76 ± 0.02	99.70 ± 0.03	**99.93 ± 0.05**
K × 100	98.74 ± 0.26	98.85 ± 0.40	99.72 ± 0.06	99.18 ± 0.21	93.26 ± 6.73	99.35 ± 0.08	99.74 ± 0.17	99.66 ± 0.04	99.60 ± 0.04	**99.89 ± 0.06**

The bold represents the best accuracy, and the underline represents the second-best accuracy.

TABLE VI
CONSUMPTION AND COMPUTATIONAL COMPLEXITY OF ALL METHODS ON EACH DATASET

Methods	**TR**					**AU**				
	Train (s)	**Test (s)**	**Params (M)**	**FLOPs (G)**	**OA (%)**	**Train (s)**	**Test (s)**	**Params (M)**	**FLOPs (G)**	**OA (%)**
LiEtAl	4.56	0.35	0.04	0.04	98.10 ± 0.24	12.95	0.84	0.04	0.04	94.84 ± 0.39
SpectralFormer	14.09	1.24	0.10	0.19	98.21 ± 0.21	59.74	3.20	0.10	0.19	94.26 ± 0.47
SSFTT	8.68	0.74	0.15	0.45	99.43 ± 0.09	31.45	1.81	0.15	0.45	97.08 ± 0.14
morpFormer	41.56	4.11	0.06	0.33	99.27 ± 0.04	191.17	10.48	0.06	0.33	96.63 ± 0.19
MDL-Cross	53.33	5.71	0.10	0.34	98.15 ± 0.46	245.29	14.40	0.10	0.34	90.03 ± 8.85
MFT_PT	83.01	12.96	0.22	0.31	98.97 ± 0.13	398.25	33.45	0.22	0.31	96.34 ± 0.10
MFT_CT	82.35	12.72	0.22	0.31	99.32 ± 0.08	386.84	32.56	0.22	0.31	96.03 ± 0.30
HCT	54.74	9.97	0.47	0.52	99.56 ± 0.09	263.70	26.94	0.73	0.57	96.89 ± 0.21
TMCFN	415.80	5.40	0.72	0.70	99.39 ± 0.02	436.80	5.40	0.71	0.94	95.29 ± 0.64
CITNet	146.97	23.48	0.56	0.63	**99.76 ± 0.04**	568.06	53.24	0.38	0.59	**97.40 ± 0.09**
Methods	**MU**					**HU**				
	Train (s)	**Test (s)**	**Params (M)**	**FLOPs (G)**	**OA (%)**	**Train (s)**	**Test (s)**	**Params (M)**	**FLOPs (G)**	**OA (%)**
LiEtAl	6.85	0.56	0.07	0.04	82.88 ± 2.30	13.19	0.24	0.09	0.04	98.84 ± 0.24
SpectralFormer	26.03	2.30	0.10	0.19	86.37 ± 0.59	46.36	0.53	0.10	0.19	98.94 ± 0.37
SSFTT	14.58	1.25	0.15	0.45	89.54 ± 0.43	28.19	0.32	0.15	0.45	99.74 ± 0.05
morpFormer	80.38	7.26	0.06	0.33	85.45 ± 0.93	139.61	1.70	0.06	0.33	99.25 ± 0.20
MDL-Cross	104.24	10.18	0.10	0.34	78.35 ± 4.84	180.50	2.47	0.10	0.34	93.79 ± 6.19
MFT_PT	174.85	24.98	0.22	0.31	86.15 ± 0.39	290.65	5.54	0.22	0.31	99.40 ± 0.08
MFT_CT	163.07	22.25	0.22	0.31	88.25 ± 0.74	291.33	5.52	0.22	0.31	99.76 ± 0.15
HCT	109.50	18.44	0.73	0.57	88.25 ± 0.74	193.89	3.95	0.73	0.57	99.69 ± 0.03
TMCFN	419.40	4.80	0.72	0.65	89.32 ± 0.12	487.80	5.95	0.76	1.15	99.63 ± 0.04
CITNet	288.21	40.43	0.92	0.69	**91.06 ± 0.62**	312.02	7.56	0.27	0.56	**99.90 ± 0.07**

The bold represents optimal accuracy.

lowest total parameters and FLOPs on the TR and MU datasets. To AU and HU datasets, the total parameters of joint use MADC and OCA or all three modules are equal and the lowest, while the FLOPs are the same and lowest when only OCA or use LGTM and OCA simultaneously. When MADC, LGTM, and OCA simultaneously exist, the classification performance is the best. The findings indicate that each component contributes positively to enhancing the accuracy of the joint classification.

2) *Details of MADC:* The 1 × 1 convolutional kernels in Section III-A learn the initial feature information as a residual join and align the output feature dimensions. Detailed ablation experimental results of other convolution sub-branches in MADC under 64 batch size and 30 PCA in Table VIII, including convolutional kernels of sizes 11, 7, 5, and 3, ac operation, and DWC operation. When employing a single convolution or combining different convolutional kernels, the model demonstrates an increase in total parameters and FLOPs. Notably, when the kernel size of 5, AC, and DWC exist simultaneously, the model achieves the lowest total parameters and FLOPs. However, when MADC is without AC or DWC, the computational complexity of the model significantly increases. Likewise, using convolutional kernels of sizes 11 and 5 together, without AC and DWC, also results in higher computational complexity. AC and DWC can numerously reduce the total parameters and computational complexity of CITNet and are close to the classification effect of ordinary convolution. When the 11 and 5 kernels are combined with AC and DWC, the model achieves lower total parameters and FLOPs while delivering the highest classification accuracy. Thus, the current structure of MADC is optimal, and all components are essential for optimizing

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TABLE VII

ABLATION EXPERIMENTS OF EACH COMPONENT ON FOUR DATASETS

3D-2D Conv	MADC	LGTM		CA	OCA	TR					AU				
		MLP	LGF			OA (%)	AA (%)	K × 100	Params	FLOPs	OA (%)	AA (%)	K × 100	Params	FLOPs
✓	×	✓	×	✓	×	99.56 ± 0.09	99.32 ± 0.13	99.42 ± 0.12	**0.47 M**	**0.52 G**	96.89 ± 0.21	86.56 ± 1.05	95.53 ± 0.34	0.73 M	0.57 G
×	✓	✓	×	✓	×	99.58 ± 0.07	99.31 ± 0.14	99.44 ± 0.10	0.72 M	0.67 G	97.27 ± 0.13	88.70 ± 0.94	96.11 ± 0.19	0.46 M	0.62 G
✓	×	×	✓	✓	×	99.59 ± 0.06	99.33 ± 0.10	99.45 ± 0.08	0.80 M	0.59 G	97.17 ± 0.14	88.29 ± 0.32	95.95 ± 0.20	0.54 M	0.54 G
✓	×	✓	×	×	✓	99.35 ± 0.06	98.94 ± 0.11	99.13 ± 0.08	0.63 M	0.55 G	96.96 ± 0.03	86.20 ± 0.84	95.65 ± 0.04	0.45 M	**0.52 G**
×	✓	×	✓	✓	×	99.60 ± 0.12	99.40 ± 0.16	99.46 ± 0.16	0.73 M	0.67 G	97.23 ± 0.21	86.79 ± 0.30	96.03 ± 0.31	0.46 M	0.61 G
×	✓	✓	×	×	✓	99.64 ± 0.06	99.45 ± 0.07	99.53 ± 0.09	0.55 M	0.63 G	97.29 ± 0.06	88.40 ± 0.24	96.12 ± 0.09	**0.38 M**	0.62 G
✓	×	×	✓	×	✓	99.65 ± 0.03	99.46 ± 0.05	99.53 ± 0.04	0.64 M	0.55 G	97.21 ± 0.18	88.14 ± 0.20	96.00 ± 0.26	0.46 M	**0.52 G**
×	✓	×	✓	×	✓	**99.76 ± 0.04**	**99.63 ± 0.05**	**99.68 ± 0.05**	0.56 M	0.63 G	**97.40 ± 0.10**	**89.19 ± 0.76**	**96.27 ± 0.14**	**0.38 M**	0.59 G
3D-2D Conv	**MADC**	**LGTM**		**CA**	**OCA**	**MU**					**HU**				
		MLP	**LGF**			**OA (%)**	**AA (%)**	**K × 100**	**Params**	**FLOPs**	**OA (%)**	**AA (%)**	**K × 100**	**Params**	**FLOPs**
✓	×	✓	×	✓	×	88.25 ± 0.74	89.99 ± 0.49	84.68 ± 0.91	**0.73 M**	**0.57 G**	99.69 ± 0.03	99.76 ± 0.02	99.66 ± 0.04	0.73 M	0.57 G
×	✓	✓	×	✓	×	89.54 ± 1.17	91.05 ± 0.63	86.32 ± 1.45	1.25 M	0.77 G	99.84 ± 0.04	99.85 ± 0.04	99.83 ± 0.04	0.35 M	0.58 G
✓	×	×	✓	✓	×	87.92 ± 0.95	89.19 ± 0.49	84.17 ± 1.14	1.33 M	0.69 G	99.86 ± 0.08	99.89 ± 0.06	99.84 ± 0.08	0.43 M	0.51 G
✓	×	✓	×	×	✓	87.65 ± 0.29	89.16 ± 0.07	83.89 ± 0.35	0.99 M	0.61 G	99.84 ± 0.04	99.87 ± 0.03	99.82 ± 0.04	0.35 M	**0.49 G**
×	✓	×	✓	✓	×	89.87 ± 1.05	90.79 ± 0.10	86.74 ± 1.32	1.25 M	0.77 G	99.80 ± 0.07	99.84 ± 0.05	99.78 ± 0.08	0.35 M	0.58 G
×	✓	✓	×	×	✓	90.51 ± 0.53	91.78 ± 0.53	87.52 ± 0.69	0.91 M	0.69 G	99.52 ± 0.12	99.63 ± 0.09	99.48 ± 0.13	**0.27 M**	0.56 G
✓	×	×	✓	×	✓	83.24 ± 6.54	86.27 ± 5.21	78.41 ± 8.15	0.99 M	0.61 G	99.78 ± 0.08	99.82 ± 0.05	99.76 ± 0.09	0.35 M	**0.49 G**
×	✓	×	✓	×	✓	**91.06 ± 0.62**	**92.37 ± 0.21**	**88.23 ± 0.78**	0.92 M	0.69 G	**99.90 ± 0.07**	**99.93 ± 0.05**	**99.89 ± 0.06**	**0.27 M**	0.56 G

The bold represents optimal performance. ✓means the component exists and × means that inexistent.

TABLE VIII

DIFFERENT COMBINATIONS OF MADC ON THE TR DATASET

11	7	5	3	AC	DWC	OA (%)	AA (%)	K × 100	Params	FLOPs
✓	×	×	×	×	×	99.58 ± 0.06	99.35 ± 0.09	99.43 ± 0.09	1.62 M	27.84 G
×	✓	×	×	×	×	99.64 ± 0.09	99.40 ± 0.16	99.51 ± 0.11	0.96 M	8.58 G
×	×	✓	×	×	×	99.50 ± 0.10	99.22 ± 0.16	99.33 ± 0.14	0.75 M	3.82 G
×	×	×	✓	×	×	99.52 ± 0.14	99.26 ± 0.22	99.36 ± 0.18	0.61 M	1.34 G
✓	✓	×	×	×	×	99.62 ± 0.08	99.40 ± 0.09	99.48 ± 0.11	2.05 M	36.18 G
✓	×	✓	×	×	×	99.68 ± 0.15	99.51 ± 0.23	99.58 ± 0.20	1.84 M	31.42 G
✓	×	×	✓	×	×	99.61 ± 0.09	99.37 ± 0.21	99.47 ± 0.13	1.71 M	28.94 G
×	✓	✓	×	×	×	99.59 ± 0.15	99.40 ± 0.19	99.45 ± 0.20	1.19 M	12.16 G
×	✓	×	✓	×	×	99.56 ± 0.19	99.33 ± 0.32	99.41 ± 0.26	1.05 M	9.69 G
×	×	✓	✓	×	×	99.52 ± 0.11	99.28 ± 0.20	99.36 ± 0.15	0.84 M	4.92 G
✓	×	×	×	✓	×	99.53 ± 0.09	99.24 ± 0.07	99.37 ± 0.13	0.67 M	1.88 G
✓	×	×	×	×	✓	99.52 ± 0.26	99.20 ± 0.43	99.36 ± 0.35	0.56 M	2.97 G
✓	×	×	×	✓	✓	99.65 ± 0.11	99.45 ± 0.16	99.53 ± 0.15	0.54 M	0.45 G
✓	✓	×	×	✓	✓	99.50 ± 0.07	98.71 ± 0.84	99.34 ± 0.09	0.56 M	0.64 G
×	×	✓	×	✓	×	99.42 ± 0.34	99.08 ± 0.59	99.23 ± 0.46	0.62 M	1.23 G
×	×	✓	×	×	✓	99.64 ± 0.10	99.05 ± 0.15	99.52 ± 0.13	0.54 M	0.63 G
×	×	✓	×	✓	✓	99.57 ± 0.03	99.35 ± 0.05	99.43 ± 0.05	**0.53 M**	**0.41 G**
✓	×	✓	×	✓	×	99.57 ± 0.06	99.35 ± 0.10	99.43 ± 0.08	0.82 M	3.25 G
✓	×	✓	×	×	✓	99.57 ± 0.02	99.28 ± 0.05	99.43 ± 0.02	0.58 M	3.36 G
✓	×	×	✓	✓	✓	99.60 ± 0.06	99.37 ± 0.12	99.47 ± 0.08	0.56 M	0.61 G
×	✓	✓	×	✓	✓	99.69 ± 0.03	99.48 ± 0.04	99.59 ± 0.03	0.56 M	0.59 G
×	✓	×	✓	✓	✓	99.54 ± 0.23	99.29 ± 0.30	99.38 ± 0.31	0.56 M	0.58 G
×	×	✓	✓	✓	✓	99.62 ± 0.05	99.40 ± 0.07	99.50 ± 0.06	0.57 M	0.56 G
✓	×	✓	×	✓	✓	**99.76 ± 0.04**	**99.63 ± 0.05**	**99.68 ± 0.05**	0.56 M	0.63 G

The bold represents optimal performance. ✓means the component exists and × means that inexistent. AC is the asymmetric convolution, and DWC is the depthwise convolution.

TABLE IX

DIFFERENT SHALLOW FEATURES EXTRACTION MODULES

Module	TR			AU		
	OA (%)	AA (%)	K × 100	OA (%)	AA (%)	K × 100
Local-Global	99.66 ± 0.05	99.44 ± 0.08	99.55 ± 0.07	97.09 ± 0.04	84.17 ± 2.19	95.82 ± 0.04
MADC	**99.76 ± 0.04**	**99.63 ± 0.05**	**99.68 ± 0.05**	**97.40 ± 0.10**	**89.19 ± 0.76**	**96.27 ± 0.14**
Module	**MU**			**HU**		
	OA (%)	**AA (%)**	**K × 100**	**OA (%)**	**AA (%)**	**K × 100**
Local-Global	86.24 ± 3.05	88.50 ± 2.37	82.04 ± 3.83	99.43 ± 0.05	99.56 ± 0.03	99.38 ± 0.05
MADC	**91.06 ± 0.62**	**92.37 ± 0.21**	**88.23 ± 0.78**	**99.90 ± 0.07**	**99.93 ± 0.05**	**99.89 ± 0.06**

The bold represents optimal performance.

performance. As shown in Table IX, a comparison of MADC with the local–global feature extraction module across four datasets reveals that MADC achieves higher classification accuracy.

3) *Details of OCA:* Table X shows the effects of AC and DWC on OCA. When nonuse of the convolution operation (Conv), OA is 99.60%, AA is 99.40%, and K × 100 is 99.46, alongside total parameters and FLOPs of 0.73 M and 0.67 G. Utilizing the original convolutional kernel yields an OA of 99.35%, an AA of 98.95%, and a K × 100 of 99.13, with 13.64 M total parameters and 7.33 G FLOPs. Utilizing the AC alone yields the OA of 99.65%, AA of 99.44%, the K × 100 of 99.53, with 5.78 M total parameters and

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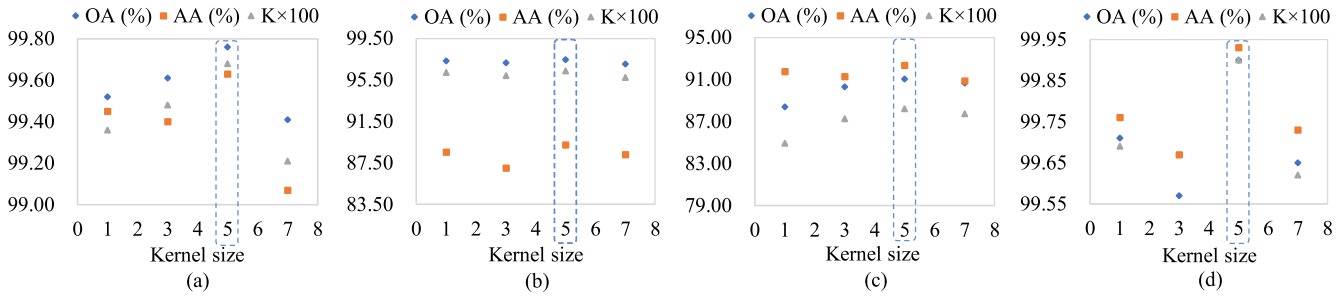


Fig. 18. Effect of the convolutional kernel size in the OCA fusion encoder on classification performance. (a) TR datset. (b) AU datset. (c) MU datset. (d) HU datset. The ordinate represents the accuracy of the three evaluation indexes, and the convolutional kernel size 5 provides the optimal classification effect.

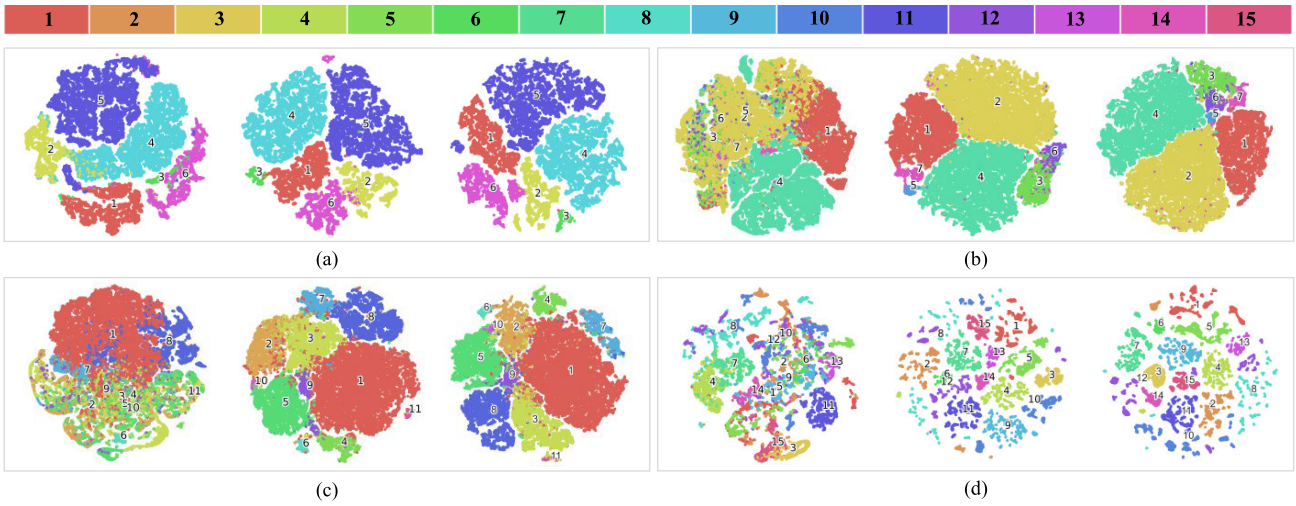


Fig. 19. T-SNE visualizations on all datasets under different input types. (a) TR dataset. (b) AU dataset. (c) MU dataset. (d) HU dataset. (Left) LiDAR-DSM, (middle) HSI, and (right) multimodal fusion data. The better clustering effect can be obtained by using multimodal data for classification.

3.29 G FLOPs. When employing the DWC alone results in an OA of 99.54%, an AA of 99.30%, and a K × 100 of 99.39, alongside total parameters and FLOPs of 0.59 M and 0.64 G. When integrating AC and DWC, the performance is enhanced, achieving an OA of 99.76%, an AA of 99.63%, and a K × 100 of 99.68, with the model complexity reduced to 0.56 M and 0.63 G. AC and DWC can improve the classification accuracy while reducing the total parameters and FLOPs. When the rank of a 2-D convolutional kernel is 1, it can be equivalent to a series of 1-D convolutions [53], which can comply with the CA structure characteristics while reducing the parameters.

4) *Convolutional Kernel Size in OCA:* This section examines the impact of convolutional kernel sizes in the OCA on fusion classification performance across various datasets, with experimental outcomes presented in Fig. 18. Due to tokens after tokenization being 4, the reconstructed spatial dimension is 2 × 2. Ding et al. [66] demonstrated that model performance can be enhanced even when the convolutional kernel size exceeds the feature map size in deep layers of the network. Motivated by this, we confidently adjust the convolutional kernel size in OCA to explore potential performance improvements. Different convolutional kernel sizes 1, 3, 5, and 7 for comparison experiments, in which the OA, AA, and K × 100 reach their optimum on all datasets when the convolutional kernel size is 5.

TABLE X
DETAILS OF OCA ON THE TR DATASET

Conv	AC	DWC	OA (%)	AA (%)	K × 100	Params	FLOPs
×	×	×	99.60 ± 0.12	99.40 ± 0.16	99.46 ± 0.16	0.73 M	0.67 G
✓	×	×	99.35 ± 0.06	98.95 ± 0.11	99.13 ± 0.08	13.64 M	7.33 G
×	✓	×	99.65 ± 0.06	99.44 ± 0.08	99.53 ± 0.08	5.78 M	3.30 G
×	×	✓	99.54 ± 0.17	99.30 ± 0.17	99.39 ± 0.23	0.59 M	0.64 G
×	✓	✓	**99.76 ± 0.04**	**99.63 ± 0.05**	**99.68 ± 0.05**	**0.56 M**	**0.63 G**

The bold represents optimal performance. ✓means the component exists and × means that inexistent.

B. Different Inputs

In this section, different inputs are compared with CITNet. From Table XI, joining HSI and LiDAR-DSM data can obtain better classification results. To the TR dataset, the integration of HSI and LiDAR-DSM data resulted in a notable improvement in classification metrics: OA increases by 0.89%, AA by 2.21%, and K × 100 by 1.18 compared with classifications performed using solely HSI. To the AU dataset, OA, AA, and K × 100 increase by 0.12%, 0.96%, and 0.16%. To the MU dataset, OA, AA, and K × 100 increase by 0.30%, 0.40%, and 0.34%. To the HU dataset, OA, AA, and K × 100 increase by 0.11%, 0.10%, and 0.12. The analysis demonstrates that the elevation features sourced from LiDAR-DSM serve as a beneficial complement to HSI, offering additional learnable information. As depicted in Fig. 19, a t-distributed stochastic neighbor embedding (T-SNE) visualization highlights a clear linkage between the accuracy and the clustering effect of the CITNet, indicating that better accuracy leads to superior

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TABLE XI
ABLATION EXPERIMENTS OF DIFFERENT INPUTS

Inputs	TR			AU		
	OA (%)	AA (%)	K × 100	OA (%)	AA (%)	K × 100
HSI	98.87 ± 0.42	97.42 ± 0.47	98.50 ± 0.56	97.28 ± 0.11	88.23 ± 1.13	96.11 ± 0.15
LiDAR-DSM	93.41 ± 2.22	81.64 ± 6.54	91.12 ± 3.12	82.40 ± 0.68	43.31 ± 0.87	73.59 ± 1.05
HSI + LiDAR-DSM	**99.76 ± 0.04**	**99.63 ± 0.05**	**99.68 ± 0.05**	**97.40 ± 0.10**	**89.19 ± 0.76**	**96.27 ± 0.14**
Inputs	**MU**			**HU**		
	OA (%)	**AA (%)**	**K × 100**	**OA (%)**	**AA (%)**	**K × 100**
HSI	90.76 ± 0.51	91.97 ± 0.64	87.89 ± 0.62	99.79 ± 0.07	99.83 ± 0.04	99.77 ± 0.08
LiDAR-DSM	54.16 ± 3.15	55.09 ± 3.66	43.67 ± 2.94	69.64 ± 3.05	72.97 ± 2.99	67.24 ± 3.26
HSI + LiDAR-DSM	**91.06 ± 0.62**	**92.37 ± 0.21**	**88.23 ± 0.78**	**99.90 ± 0.07**	**99.93 ± 0.05**	**99.89 ± 0.06**

The bold represents the best accuracy.

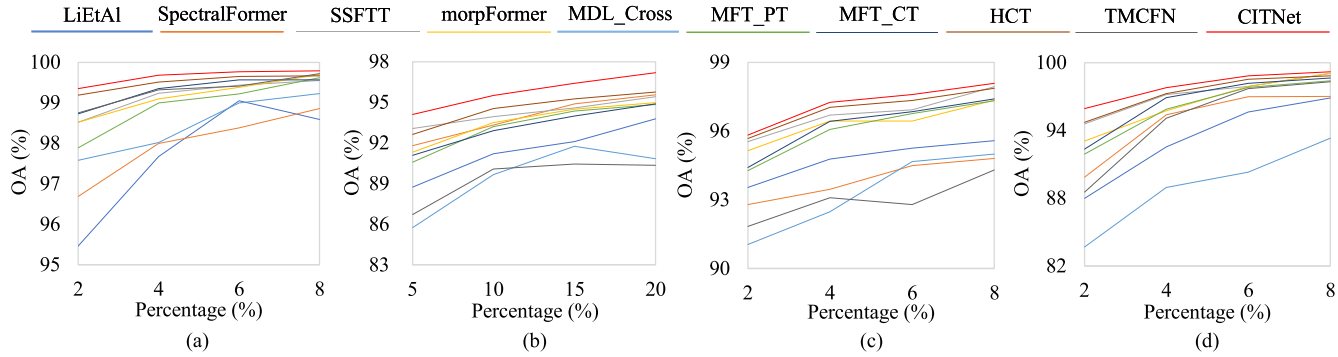


Fig. 20. Classification results of different training samples. (a) TR dataset. (b) AU dataset. (c) MU dataset. (d) HU dataset. The accuracies of all methods significantly improved with an increase in training samples. The CITNet model consistently demonstrated superior performance across all scenarios.

clustering performance. The classification result visualization is shown on the left for the LiDAR-DSM input, in the middle for the HSI input, and on the right for the combined input. The optimal clustering effectiveness is attained by joint HSI and LiDAR-DSM data, underscoring the significant advantage of multimodal data in enhancing the model's classification accuracy.

C. Training Samples

In this section, experiments analyzed the performance of the proposed CITNet under different training samples. Experimental settings are the same as above, and results in Fig. 20. To TR, AU, and HU datasets, 2%, 4%, 6%, and 8% of total samples are selected as the training set, respectively. Considering the significantly uneven sample distribution within the MU dataset, we select 5%, 10%, 15%, and 20% of the total samples for training. The experimental results demonstrate a significant enhancement in the accuracies of all methods as the number of training samples increased. Among them, the CITNet model consistently outperformed other methods across various scenarios, especially showing a marked improvement in accuracy on the MU dataset. This superior performance is attributed to the local–global feature extraction of CITNet and the spatial information modeling facilitated by OCA, enabling it to adapt more effectively to uneven distributions and improve classification accuracy. Furthermore, the effectiveness of CITNet across various datasets indicates its broad applicability in tasks of multimodal data joint classification.

VI. CONCLUSION

The study explores a novel convolution interaction transformer model based on local–global feature dependency for the joint classification of HSI and LiDAR-DSM data. The proposed model effectively merges CNN and the transformer to proficiently capture and integrate spectral–spatial features from HSI and elevation information from LiDAR-DSM data. The MADC component adequately learns local–global correlations within shallow features through a multiscale CNN approach. Meanwhile, the LGTM extracts local–global features by interacting with spatial and channel features. The OCA mechanism effectively models the spatial relationships of semantic tokens, thereby enhancing the joint classification performance. In conclusion, CITNet adeptly integrates local–global features from multimodal data and significantly advances the joint classification outcomes for HSI and LiDAR-DSM data.

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